



Low Regularity Integrators for the Conservative Allen–Cahn Equation with a Nonlocal Constraint

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Abstract

In contrast to the classical Allen–Cahn equation, the conservative Allen–Cahn equation with a nonlocal Lagrange multiplier not only satisfies the maximum bound principle (MBP) and energy dissipation law but also ensures mass conservation. Many existing schemes often fail to preserve all these properties at the discrete level or require high regularity in time on the exact solution for convergence analysis. In this paper, we construct a new class of low regularity integrators (LRIs) for time discretization of the conservative Allen–Cahn equation by repeatedly using Duhamel’s formula. The proposed first- and second-order LRI schemes are shown to conserve mass unconditionally and satisfy the MBP under some time step size constraints. Temporal error estimates for these schemes are derived under a low regularity assumption that the exact solution is only Lipschitz continuous in time, followed by a rigorous proof for energy stability of the corresponding time-discrete solutions. Various numerical experiments and comparisons in two and three dimensions are presented to verify the theoretical results and illustrate the performance of the LRI schemes, especially when the interfacial parameter approaches zero.

Keywords Conservative Allen–Cahn equation · Low regularity integrators · Mass conservation · Maximum bound principle · Energy stability

Mathematics Subject Classification 35K55 · 35B50 · 65M12 · 65R20

1 Introduction

The Allen–Cahn equation, initially proposed by Allen and Cahn [2], describes the phase separation process of binary alloys at a specific temperature. It is a well-known L^2 gradient

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flow with respect to the Ginzburg–Landau free energy

$$E(u) = \int_{\Omega} \left(\frac{\varepsilon^2}{2} |\nabla u|^2 + F(u) \right) dx,$$

and is given by

$$u_t = -\frac{\delta E}{\delta u} = \varepsilon^2 \Delta u + f(u), \quad x \in \Omega, \quad t > 0,$$

where $\frac{\delta E}{\delta u}$ denotes the variational derivative of $E(u)$, $u(x, t)$ is the unknown function, $\Omega \subset \mathbb{R}^d$ ($d \leq 3$) is a bounded domain with Lipschitz boundary, $\varepsilon > 0$ is a given parameter representing the width of diffuse interface, f is a real-valued continuous nonlinear function, and F is an energy potential with $F' = -f$. An important property of the solution to the Allen–Cahn equation is the maximum bound principle (MBP) [37], in the sense that the solution is uniformly bounded for all time by a constant imposed by the initial data under certain assumptions on the nonlinear function. Although the classical Allen–Cahn equation satisfies the MBP, it fails to preserve the total mass over time. The Cahn–Hilliard equation, on the other hand, naturally conserves the mass of material components [9, 10] but does not satisfy the MBP due to the fourth-order spatial derivatives, making it computationally challenging to solve. Therefore, it is necessary to consider the conservative Allen–Cahn equation [5, 31, 32], where an additional nonlocal Lagrange multiplier is introduced to conserve the total mass while still preserving the MBP. The conservative Allen–Cahn equation, similar to its counterpart, holds great significance in a wide range of applications such as modeling two-phase fluid flows [38, 55] and multiphase fluid flows [1, 23, 49], solving PDEs on surfaces [27, 56], and examining diblock copolymers [17]. In this paper, we are interested in the following conservative Allen–Cahn equation:

$$u_t = \varepsilon^2 \Delta u + \bar{f}(u), \quad x \in \Omega, \quad t > 0, \quad (1.1)$$

subject to the initial condition $u(\cdot, 0) = u_0$ and the periodic or homogeneous Neumann boundary conditions, where $\bar{f}(u)$ is the modified nonlinear function containing a nonlocal term, introduced by Rubinstein and Sternberg [40] and defined as

$$\bar{f}(u)(x, t) = f(u(x, t)) - \frac{1}{|\Omega|} \int_{\Omega} f(u(y, t)) dy. \quad (1.2)$$

It is well-known that the exact solution to the conservative Allen–Cahn equation (1.1)–(1.2) under the periodic or homogeneous Neumann boundary conditions decreases the energy functional and conserves mass along the time:

$$\frac{d}{dt} E(u) = - \int_{\Omega} |u_t|^2 dx \leq 0, \quad \frac{d}{dt} \int_{\Omega} u(x, t) dx = 0.$$

Other versions of the conservative Allen–Cahn equation can be found in [3, 7, 22, 24, 28, 58], where different formulations for $\bar{f}$ are used instead of the space-independent Lagrange multiplier in (1.2); however, such equations only conserve mass but do not satisfy the energy dissipation law.

Due to the importance of energy stability in phase field models, various energy stable schemes have been extensively investigated. Examples of traditional approaches include the implicit methods [15, 53], the convex splitting schemes [45, 48], the linear stabilized semi-implicit methods [44, 47], while the invariant energy quadratization [54, 57] and the scalar

auxiliary variable [6, 42] are among the advanced techniques that have been widely applied in gradient flow problems.

The development of MBP-preserving numerical methods for Allen–Cahn type equations has received significant attention recently. In [8], the first-order convex splitting scheme for the Allen–Cahn equation was shown to be unconditionally energy stable, followed by the pointwise boundedness of the numerical solution. Later in [47], a MBP-preserving scheme was developed by combining the standard implicit-explicit scheme in time and the central finite difference method in space. The generalized Allen–Cahn equation with a nonlinear degenerated mobility was studied in [43], where the semi-implicit scheme preserves the MBP under certain time step constraints. Following these ideas, several MBP-preserving schemes have been developed, including Crank–Nicolson/Adams–Bashforth schemes [19, 20, 46], a new second-order finite difference scheme with proper parameters [21], and the modified leap-frog scheme with a stabilized term [16]. These schemes not only ensure the preservation of the MBP but also guarantee energy stability for the Allen–Cahn equation and its variants. Moreover, several MBP-preserving schemes have been proposed for the surface Allen–Cahn equation [52] and the fractional/nonlocal Allen–Cahn equations [13, 18, 34, 59]. Despite considerable progress in MBP-preserving schemes, existing numerical methods for the conservative Allen–Cahn equation are still limited. Operator splitting methods have been proved to unconditionally preserve the MBP by introducing a different Lagrange multiplier, although obtaining energy dissipation properties remains challenging [28–30, 58]. In [32], the first- and second-order stabilized exponential time differencing (ETD1 and ETD2) schemes were proved to unconditionally preserve the MBP. The ETD1 method was shown to satisfy the energy decay property while the ETD2 method provided a uniform bound for the energy functional. In [60], the improved stabilized integrating factor Runge–Kutta (isIFRK) schemes up to third-order were studied for two types of the conservative Allen–Cahn equation. The isIFRK schemes not only preserve the MBP but also conserve mass to machine accuracy while the original stabilized integrating factor Runge–Kutta schemes fail to preserve mass unless the stabilizing constant and the initial mass satisfy some specific conditions.

It should be emphasized that many existing schemes, such as ETD or isIFRK, require high regularity in time on the exact solution for their convergence analysis, with the solution being at least in C^p for the p th-order scheme to converge. Recently, a new class of schemes known as low regularity integrators (LRIs) has been introduced to remove such a requirement on the exact solution. In [4, 35, 36, 41, 50, 51], the so-called resonance-based schemes were developed to control nonlinear terms at low regularity in dispersive equations based on Fourier series expansion of the exact solution. Since this approach requires periodic boundary conditions, a general framework of LRI schemes [39] was proposed for evolution PDEs based on Duhamel’s principle, in which different types of boundary conditions can be considered. A combination of LRI and the finite element method was established for approximating the solution of the Navier–Stokes equations [33], where the first-order convergence of the method is guaranteed only if the spatial mesh size is smaller than the time step size up to some constant. In addition to the finite element approach, LRI schemes can be coupled with the central finite difference method [11] for semilinear parabolic equations and were shown to preserve the MBP conditionally as well as satisfy energy stability. Fully discrete convergence analysis of the proposed first-order schemes is given in [12] assuming that the exact solution is continuous in time.

The goal of this paper is to develop and analyze first- and second-order LRI schemes for time discretization of the conservative Allen–Cahn equation (1.1). Due to the presence of the nonlocal Lagrange term defined in (1.2), the derivation and analysis of LRIs, especially the second-order scheme, are substantially different and more challenging than for the case with

the classical Allen–Cahn equation as studied in [11, 12]. In addition, in this work we mainly focus on analyzing the time discretization of the PDE, noting that LRIs can be coupled with various spatial discretizations, including finite difference, finite volume and lumped-mass finite element methods [14]. The proposed LRI schemes are shown to unconditionally conserve mass and preserve the MBP under some time step size requirements. In comparison to the conditional MBP for the classical Allen–Cahn equation [11], these conditions on the time step size are more restrictive because of the nonlocal Lagrange term. The convergence analysis of the LRI schemes is carried out under the assumption that the exact solution is Lipschitz continuous in time. By exploiting the commutativity of Laplacian exponential and spatial differential operators in rectangular domains with the periodic boundary conditions, we demonstrate the energy stability of the proposed schemes through the utilization of both MBP and error estimate results. Numerical experiments show the outperformance of LRIs compared to the ETD methods of the same order, especially when ε approaches zero. Moreover, all the LRI schemes preserve the three intrinsic properties of the conservative Allen–Cahn equation, namely mass conservation, MBP and energy stability. To the best of our knowledge, this is the first time that LRI schemes are studied for a PDE containing nonlocal terms.

The rest of the paper is organized as follows: in Sect. 2, we construct the first- and second-order LRI schemes for the model problem (1.1)–(1.2). Section 3 focuses on mass conservation and the MBP of the time-discrete LRI solution. Convergence analysis and energy stability are proved in Sects. 4 and 5, respectively. To verify the theoretical results and demonstrate the performance of the proposed LRIs, various two and three dimensional numerical experiments are presented in Sect. 6. Finally, some concluding remarks are given in Sect. 7.

2 Low Regularity Integrator Schemes

For a fixed $T > 0$, consider a uniform partition of the time interval $[0, T]$ with $0 = t_0 < t_1 < \dots < t_M = T$ and the time step size $\tau := T/M$. Throughout the paper, we write $u(t)$ in place of $u(\cdot, t)$ and denote by u^m the approximation of $u(t_m)$. We also write $\int_{\Omega} v$ instead of $\int_{\Omega} v(y) dy$ when there is no confusion. By applying Duhamel's principle to (1.1), there holds

$$u(t_{m+1}) = e^{\tau\varepsilon^2\Delta}u(t_m) + \int_0^\tau e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(u(t_m+s))ds. \quad (2.1)$$

We also have from (1.1) and Duhamel's principle that

$$u(t_m+s) = e^{s\varepsilon^2\Delta}u(t_m) + \int_0^s e^{(s-\sigma)\varepsilon^2\Delta}\bar{f}(u(t_m+\sigma))d\sigma. \quad (2.2)$$

By approximating $u(t_m+s)$ in (2.1) by the first term $e^{s\varepsilon^2\Delta}u(t_m)$ in (2.2) and using one-point quadrature rules as in [11], we obtain the following first-order LRI schemes, namely the LRIIa scheme:

$$u^{m+1} = e^{\tau\varepsilon^2\Delta}(u^m + \tau\bar{f}(u^m)), \quad (2.3)$$

and the LRIIb scheme:

$$u^{m+1} = e^{\tau\varepsilon^2\Delta}u^m + \tau\bar{f}(e^{\tau\varepsilon^2\Delta}u^m). \quad (2.4)$$

To construct a second-order LRI scheme, we assume that f is a C^1 function. First we approximate $u(t_m + s)$ in (2.1) by $e^{s\varepsilon^2\Delta}u(t_m) + s\bar{f}(u(t_m))$ from (2.2) and obtain

$$u^{m+1} \approx e^{\tau\varepsilon^2\Delta}u^m + \int_0^\tau e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(e^{s\varepsilon^2\Delta}u^m + s\bar{f}(u^m)) ds. \quad (2.5)$$

Then by the definition of $\bar{f}$ and Taylor expansion, we observe that

$$\begin{aligned} \bar{f}(e^{s\varepsilon^2\Delta}u^m + s\bar{f}(u^m)) &= f(e^{s\varepsilon^2\Delta}u^m + s\bar{f}(u^m)) - \frac{1}{|\Omega|} \int_\Omega f(e^{s\varepsilon^2\Delta}u^m + s\bar{f}(u^m)) \\ &\approx f(e^{s\varepsilon^2\Delta}u^m) + sf'(e^{s\varepsilon^2\Delta}u^m)\bar{f}(u^m) - \frac{1}{|\Omega|} \int_\Omega \left(f(e^{s\varepsilon^2\Delta}u^m) + sf'(e^{s\varepsilon^2\Delta}u^m)\bar{f}(u^m) \right) \\ &= \bar{f}(e^{s\varepsilon^2\Delta}u^m) + s \left[f'(e^{s\varepsilon^2\Delta}u^m)\bar{f}(u^m) - \frac{1}{|\Omega|} \int_\Omega f'(e^{s\varepsilon^2\Delta}u^m)\bar{f}(u^m) \right]. \end{aligned} \quad (2.6)$$

Combining (2.5) and (2.6) yields

$$\begin{aligned} u^{m+1} &\approx e^{\tau\varepsilon^2\Delta}u^m + \int_0^\tau e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(e^{s\varepsilon^2\Delta}u^m) ds \\ &\quad + \int_0^\tau se^{(\tau-s)\varepsilon^2\Delta} \left[f'(e^{s\varepsilon^2\Delta}u^m)\bar{f}(u^m) - \frac{1}{|\Omega|} \int_\Omega f'(e^{s\varepsilon^2\Delta}u^m)\bar{f}(u^m) \right] ds. \end{aligned}$$

By using the trapezoidal rule for the second term and setting $s = 0$ in all exponents of the third term on the right hand side of the above approximation, the second-order LRI scheme, namely the LRI2 scheme, is given by

$$u^{m+1} = e^{\tau\varepsilon^2\Delta}u^m + \frac{\tau}{2} \left[e^{\tau\varepsilon^2\Delta}\bar{f}(u^m) + \bar{f}(e^{\tau\varepsilon^2\Delta}u^m) \right] + \frac{\tau^2}{2} e^{\tau\varepsilon^2\Delta}\bar{g}(u^m). \quad (2.7)$$

Here $\bar{g}$ is defined as $\bar{g}(u^m) = g(u^m) - \frac{1}{|\Omega|} \int_\Omega g(u^m)$, where $g := f'f$.

We remark that, due to the nonlocal Lagrange term in $\bar{f}$, the formulations of the LRI schemes, especially LRI2, for the conservative Allen–Cahn equation are more complicated than those for the classical Allen–Cahn equation [11]. Consequently, the analysis presented in the following is more challenging and significantly different from [11, 12].

3 Mass Conservation and Discrete MBP

We demonstrate in this section the unconditional mass conservation and the conditional MBP of the first- and second-order LRI schemes. The MBP results are of significant importance for our subsequent analysis.

3.1 Mass Conservation

For any two functions $v_1, v_2 \in L^2(\Omega)$, we denote by $(v_1, v_2) = \int_\Omega v_1(x)v_2(x) dx$ the standard L^2 inner product of v_1 and v_2 . The lemma below states a fundamental characteristic of the heat equation under suitable boundary conditions.

Lemma 3.1 For any $v \in L^2(\Omega)$, the exponential of the Laplace operator with the periodic or homogeneous Neumann boundary conditions has the following property

$$(e^{t\Delta}v, 1) = (v, 1), \quad \forall t > 0.$$

Proof Fix $v \in L^2(\Omega)$ and consider the following heat equation:

$$\begin{cases} w_t = \Delta w, & \mathbf{x} \in \Omega, t > 0, \\ w(\mathbf{x}, 0) = v(\mathbf{x}), & \mathbf{x} \in \overline{\Omega}, \end{cases}$$

subject to the periodic or homogeneous Neumann boundary conditions. Taking the L^2 inner product with 1 on both sides of the above equation yields

$$\frac{d}{dt}(w, 1) = (\Delta w, 1) = 0, \quad t > 0.$$

This means that $(w(t), 1)$ is a constant for any $t > 0$. Since $w(t) \rightarrow w(0) = v$ almost everywhere as $t \rightarrow 0^+$, we deduce that $(w(t), 1) = (v, 1)$ for all $t > 0$, which completes the proof with the notice that $w(t) = e^{t\Delta}w(0) = e^{t\Delta}v$. $\square$

Theorem 3.1 The first-order LRI schemes (2.3) and (2.4) conserve mass unconditionally, i.e., for any time step size $\tau > 0$, we have

$$(u^m, 1) = (u^0, 1), \quad m = 1, 2, \dots, M.$$

Proof We first consider the LRI1a scheme (2.3). By taking the L^2 inner product with 1 on both sides of (2.3), we arrive at

$$(u^{m+1}, 1) = (e^{\tau\varepsilon^2\Delta}(u^m + \tau\bar{f}(u^m)), 1). \quad (3.1)$$

From Lemma 3.1 and (3.1), we have

$$(u^{m+1}, 1) = (u^m + \tau\bar{f}(u^m), 1) = (u^m, 1),$$

since $(\bar{f}(u^m), 1) = 0$ (cf. (1.2)). Similarly, from the LRI1b scheme (2.4) and Lemma 3.1, we obtain

$$\begin{aligned} (u^{m+1}, 1) &= (e^{\tau\varepsilon^2\Delta}u^m + \tau\bar{f}(e^{\tau\varepsilon^2\Delta}u^m), 1) \\ &= (e^{\tau\varepsilon^2\Delta}u^m, 1) + \tau(\bar{f}(e^{\tau\varepsilon^2\Delta}u^m), 1) = (u^m, 1), \end{aligned}$$

which concludes the proof. $\square$

Theorem 3.2 The LRI2 scheme (2.7) conserves mass unconditionally, i.e., for any time step size $\tau > 0$, we have

$$(u^m, 1) = (u^0, 1), \quad m = 1, 2, \dots, M.$$

Proof Taking the L^2 inner product with 1 on both sides of (2.7) yields

$$\begin{aligned} (u^{m+1}, 1) &= \frac{1}{2}(e^{\tau\varepsilon^2\Delta}(u^m + \tau\bar{f}(u^m)), 1) + \frac{1}{2}(e^{\tau\varepsilon^2\Delta}u^m + \tau\bar{f}(e^{\tau\varepsilon^2\Delta}u^m), 1) \\ &\quad + \frac{\tau^2}{2}(e^{\tau\varepsilon^2\Delta}\bar{g}(u^m), 1). \end{aligned} \quad (3.2)$$

Based on the proof of Theorem 3.1, we observe that

$$(e^{\tau\varepsilon^2\Delta}(u^m + \tau\bar{f}(u^m)), 1) = (u^m, 1) = (e^{\tau\varepsilon^2\Delta}u^m + \tau\bar{f}(e^{\tau\varepsilon^2\Delta}u^m), 1). \quad (3.3)$$

We now apply Lemma 3.1 with $v = \bar{g}(u^m)$ to obtain

$$(e^{\tau \varepsilon^2 \Delta} \bar{g}(u^m), 1) = (\bar{g}(u^m), 1) = 0. \quad (3.4)$$

Combining (3.2), (3.3) and (3.4) gives us

$$(u^{m+1}, 1) = \frac{1}{2}(u^m, 1) + \frac{1}{2}(u^m, 1) = (u^m, 1).$$

The proof is thus completed. $\square$

3.2 Discrete MBP

We first present some preliminary results that are necessary to prove the discrete MBP as well as the error estimates and energy stability in the next sections. The following property of the Laplace operator from [32, Lemma 2.1] plays a crucial role in our analysis throughout the paper.

Lemma 3.2 *The Laplace operator with the periodic or homogeneous Neumann boundary conditions generates a contraction semigroup $\{e^{t\Delta}\}_{t \geq 0}$ with respect to the supremum norm on $C(\overline{\Omega})$. Moreover,*

$$\|e^{t\Delta} v\| \leq \|v\|, \quad \forall t \geq 0, v \in C(\overline{\Omega}),$$

where $\|v\| = \max_{x \in \overline{\Omega}} |v(x)|$.

Similar to [32, 60], we impose the following assumption on f :

(A1) There exists $\beta > 0$ such that $f(\beta) \leq f(w) \leq f(-\beta)$ for all $w \in [-\beta, \beta]$.

Under Assumption (A1) and Lemma 3.2, solution to the original equation (1.1) satisfies the MBP as presented in [32, Theorem 3.1]. Let us denote $\Omega_T = \Omega \times (0, T)$.

Lemma 3.3 *If f satisfies Assumption (A1) and $u_0 \in C(\overline{\Omega})$ with $\|u_0\| \leq \beta$, then the conservative Allen–Cahn equation (1.1) admits a unique solution $u \in C(\overline{\Omega_T})$, and moreover, $\|u(t)\| \leq \beta$ for all $t \in [0, T]$.*

In order to obtain the discrete MBP results, for the first-order LRI schemes we further assume that

(A2) f is continuously differentiable and nonconstant on $[-\beta, \beta]$.

For the second-order LRI scheme, a higher regularity assumption on f is required as follows:

(A3) f is twice continuously differentiable and nonconstant on $[-\beta, \beta]$.

When (A1) and (A2) hold, we define the constants $\tau_0 := \max_{|\xi| \leq \beta} f'(\xi)$ and $\tau_1 := -(\min_{|\xi| \leq \beta} f'(\xi))^{-1}$. Clearly, $\tau_1 > 0$. In addition, whenever $f \in C^k([-\beta, \beta])$ for some non-negative integer k , we denote by

$$\alpha_k := \max_{|\xi| \leq \beta} |f^{(k)}(\xi)|. \quad (3.5)$$

The technical lemma below is fundamental in establishing the MBP for LRI schemes.

Lemma 3.4 *If f satisfies Assumption (A1), then for any $v \in C(\overline{\Omega})$ with $\|v\| \leq \beta$, we have*

$$\|v + \omega \bar{f}(v)\| \leq \beta, \quad \forall \omega \in (0, \tau_1].$$

Proof Denote by $f_0(\xi) = \xi + \omega f(\xi)$ for $\xi \in \mathbb{R}$. If $|\xi| \leq \beta$ then for all $\omega \in (0, \tau_1]$, we find that

$$f'_0(\xi) = 1 + \omega f'(\xi) \geq 1 + \omega \min_{|\xi| \leq \beta} f'(\xi) \geq 1 + \tau_1 \min_{|\xi| \leq \beta} f'(\xi) = 0.$$

This shows that f_0 is non-decreasing on $[-\beta, \beta]$, which implies

$$-\beta + \omega f(-\beta) \leq f_0(\xi) \leq \beta + \omega f(\beta), \quad \forall \xi \in [-\beta, \beta].$$

Since $\|v\| \leq \beta$, that is, $|v(\mathbf{x})| \leq \beta$ for all $\mathbf{x} \in \overline{\Omega}$, we obtain

$$-\beta + \omega f(-\beta) \leq f_0(v(\mathbf{x})) \leq \beta + \omega f(\beta), \quad \forall \mathbf{x} \in \overline{\Omega}. \quad (3.6)$$

On the other hand, by the mean value theorem and Assumption (A1), we have

$$f(\beta) \leq \frac{1}{|\Omega|} \int_{\Omega} f(v) \leq f(-\beta). \quad (3.7)$$

Combining (3.6) and (3.7) yields

$$-\beta \leq v(\mathbf{x}) + \omega \left(f(v(\mathbf{x})) - \frac{1}{|\Omega|} \int_{\Omega} f(v) \right) \leq \beta, \quad \forall \mathbf{x} \in \overline{\Omega}.$$

This completes the proof of the lemma. $\square$

Theorem 3.3 Suppose that $u_0 \in C(\overline{\Omega})$ with $\|u_0\| \leq \beta$ and f satisfies Assumptions (A1) and (A2), then the numerical solution $\{u^m\}_{0 \leq m \leq M}$ given by the LRI1a scheme (2.3) or the LRI1b scheme (2.4) is also bounded by β in the supremum norm:

$$\|u^m\| \leq \beta, \quad m = 0, 1, \dots, M,$$

provided that $0 < \tau \leq \tau_1$.

Proof It suffices to show that if $\|u^m\| \leq \beta$ then $\|u^{m+1}\| \leq \beta$. Indeed, by taking the supremum norm on both sides of the LRI1a scheme, we deduce from Lemmas 3.2 and 3.4 that

$$\|u^{m+1}\| = \|e^{\tau \varepsilon^2 \Delta} (u^m + \tau \bar{f}(u^m))\| \leq \|u^m + \tau \bar{f}(u^m)\| \leq \beta.$$

Similarly, regarding the LRI1b scheme, we have

$$\|u^{m+1}\| = \|e^{\tau \varepsilon^2 \Delta} u^m + \tau \bar{f}(e^{\tau \varepsilon^2 \Delta} u^m)\| \leq \beta,$$

because of Lemma 3.4 and the fact that $\|e^{\tau \varepsilon^2 \Delta} u^m\| \leq \|u^m\| \leq \beta$. $\square$

The LRI2 scheme also preserves the MBP conditionally as stated in the next theorem. Note that under Assumption (A3), the constant α_2 (cf. (3.5)) is well-defined. Moreover, let $\delta_0 \in (0, \frac{7}{8})$ and $\tau_2 > 0$ be given by

$$\delta_0 := \frac{1}{2\tau_1} \sqrt{\frac{\frac{7}{4} - 2\delta_0}{\alpha_2(f(-\beta) - f(\beta))}},$$

$$\tau_2 := \min \left\{ 2\delta_0\tau_1, \left(\frac{1}{\tau_1} + \tau_0 - f'(\beta) \right)^{-1}, \left(\frac{1}{\tau_1} + \tau_0 - f'(-\beta) \right)^{-1} \right\}.$$

Theorem 3.4 Suppose that $u_0 \in C(\overline{\Omega})$ with $\|u_0\| \leq \beta$ and f satisfies Assumptions (A1) and (A3), then the numerical solution $\{u^m\}_{0 \leq m \leq M}$ given by the LRI2 scheme (2.7) is also bounded by β in the supremum norm:

$$\|u^m\| \leq \beta, \quad m = 0, 1, \dots, M,$$

provided that $0 < \tau \leq \tau_2$.

Proof Assume that $\|u^m\| \leq \beta$ for some m , we need to prove that $\|u^{m+1}\| \leq \beta$. By (2.7) and the triangle inequality, we have

$$\begin{aligned} \|u^{m+1}\| &= \left\| e^{\tau \varepsilon^2 \Delta} u^m + \frac{\tau}{2} \left[e^{\tau \varepsilon^2 \Delta} \bar{f}(u^m) + \bar{f}(e^{\tau \varepsilon^2 \Delta} u^m) \right] + \frac{\tau^2}{2} e^{\tau \varepsilon^2 \Delta} \bar{g}(u^m) \right\| \\ &\leq \left\| \delta e^{\tau \varepsilon^2 \Delta} u^m + \frac{\tau}{2} \bar{f}(e^{\tau \varepsilon^2 \Delta} u^m) \right\| \\ &\quad + \left\| (1 - \delta) e^{\tau \varepsilon^2 \Delta} u^m + \frac{\tau}{2} e^{\tau \varepsilon^2 \Delta} \bar{f}(u^m) + \frac{\tau^2}{2} e^{\tau \varepsilon^2 \Delta} \bar{g}(u^m) \right\|, \end{aligned}$$

for some $\delta \in (0, 1)$ to be determined later. By Lemma 3.2, we arrive at

$$\|u^{m+1}\| \leq \delta \left\| e^{\tau \varepsilon^2 \Delta} u^m + \frac{\tau}{2\delta} \bar{f}(e^{\tau \varepsilon^2 \Delta} u^m) \right\| + \left\| (1 - \delta) u^m + \frac{\tau}{2} \bar{f}(u^m) + \frac{\tau^2}{2} \bar{g}(u^m) \right\|. \quad (3.8)$$

Since $\|u^m\| \leq \beta$, we have $\|e^{\tau \varepsilon^2 \Delta} u^m\| \leq \beta$. So if $\frac{\tau}{2\delta} \leq \tau_1$, or $\tau \leq 2\delta\tau_1$, then Lemma 3.4 gives us

$$\left\| e^{\tau \varepsilon^2 \Delta} u^m + \frac{\tau}{2\delta} \bar{f}(e^{\tau \varepsilon^2 \Delta} u^m) \right\| \leq \beta. \quad (3.9)$$

To estimate the second term on the right hand side of (3.8), let us introduce some constants:

$$\rho_0 := \frac{1}{|\Omega|} \int_{\Omega} f(u^m), \quad \rho_1 := \frac{1}{|\Omega|} \int_{\Omega} f'(u^m), \quad \rho_2 := \frac{1}{|\Omega|} \int_{\Omega} f(u^m) f'(u^m).$$

Since $\bar{f}(u^m) = f(u^m) - \rho_0$, we can rewrite $\bar{g}(u^m)$ as follows

$$\begin{aligned} \bar{g}(u^m) &= g(u^m) - \frac{1}{|\Omega|} \int_{\Omega} g(u^m) = f'(u^m) \bar{f}(u^m) - \frac{1}{|\Omega|} \int_{\Omega} f'(u^m) \bar{f}(u^m) \\ &= f'(u^m)(f(u^m) - \rho_0) - \frac{1}{|\Omega|} \int_{\Omega} f'(u^m)(f(u^m) - \rho_0) \\ &= f'(u^m)(f(u^m) - \rho_0) + \rho_0 \rho_1 - \rho_2. \end{aligned}$$

Therefore,

$$\begin{aligned} &(1 - \delta) u^m + \frac{\tau}{2} \bar{f}(u^m) + \frac{\tau^2}{2} \bar{g}(u^m) \\ &= (1 - \delta) u^m + \frac{\tau}{2} (f(u^m) - \rho_0) + \frac{\tau^2}{2} [f'(u^m)(f(u^m) - \rho_0) + \rho_0 \rho_1 - \rho_2] \quad (3.10) \\ &= (1 - \delta) u^m + \frac{\tau}{2} (f(u^m) - \rho_0) (1 + \tau f'(u^m)) + \frac{\tau^2}{2} (\rho_0 \rho_1 - \rho_2). \end{aligned}$$

The expression (3.10) motivates us to consider the function:

$$K(\xi) = (1 - \delta) \xi + \frac{\tau}{2} (f(\xi) - \rho_0) (1 + \tau f'(\xi)) + \frac{\tau^2}{2} (\rho_0 \rho_1 - \rho_2),$$

for $\xi \in [-\beta, \beta]$. Notice that

$$K'(\xi) = 1 - \delta + \frac{\tau}{2} [f'(\xi)(1 + \tau f'(\xi)) + \tau f''(\xi)(f(\xi) - \rho_0)].$$

By the mean value theorem and Assumption (A1), we find that $f(\beta) \leq \rho_0 \leq f(-\beta)$. This shows that $|f(\xi) - \rho_0| \leq f(-\beta) - f(\beta)$ for all $\xi \in [-\beta, \beta]$. Hence, we obtain

$$\begin{aligned} K'(\xi) &= \frac{7}{8} - \delta + \frac{1}{2} \left(\frac{1}{2} + \tau f'(\xi) \right)^2 + \frac{\tau^2}{2} f''(\xi)(f(\xi) - \rho_0) \\ &\geq \frac{7}{8} - \delta - \frac{\tau^2}{2} \alpha_2 (f(-\beta) - f(\beta)), \quad \forall \xi \in [-\beta, \beta]. \end{aligned}$$

If $\delta \in (0, \frac{7}{8})$ and τ are chosen such that

$$\tau \leq \sqrt{\frac{\frac{7}{4} - 2\delta}{\alpha_2 (f(-\beta) - f(\beta))}},$$

then $K'(\xi) \geq 0$ for all $\xi \in [-\beta, \beta]$. Since K is non-decreasing on $[-\beta, \beta]$, it suffices to control the values of K at the endpoints, i.e., we aim to show that $K(\beta) \leq (1 - \delta)\beta$ and $K(-\beta) \geq -(1 - \delta)\beta$. Observe that

$$\begin{aligned} K(\beta) &= (1 - \delta)\beta + \frac{\tau}{2} (f(\beta) - \rho_0)(1 + \tau f'(\beta)) + \frac{\tau^2}{2} (\rho_0 \rho_1 - \rho_2), \\ K(-\beta) &= -(1 - \delta)\beta + \frac{\tau}{2} (f(-\beta) - \rho_0)(1 + \tau f'(-\beta)) + \frac{\tau^2}{2} (\rho_0 \rho_1 - \rho_2). \end{aligned}$$

By the definition of ρ_1 and the fact that $-\frac{1}{\tau_1} \leq f'(\xi) \leq \tau_0$ for all $\xi \in [-\beta, \beta]$, we obtain

$$-\frac{1}{\tau_1} \leq \rho_1 \leq \tau_0. \text{ Hence}$$

$$\begin{aligned} \rho_0 \rho_1 - \rho_2 &= \rho_0 \rho_1 - \frac{1}{|\Omega|} \int_{\Omega} f(u^m) f'(u^m) \\ &= \rho_1 (\rho_0 - f(\beta)) - \frac{1}{|\Omega|} \int_{\Omega} [f(u^m) - f(\beta)] f'(u^m) \\ &\leq \tau_0 (\rho_0 - f(\beta)) + \frac{1}{\tau_1} \frac{1}{|\Omega|} \int_{\Omega} [f(u^m) - f(\beta)] = (\rho_0 - f(\beta)) \left(\tau_0 + \frac{1}{\tau_1} \right). \end{aligned}$$

Similarly,

$$\begin{aligned} \rho_0 \rho_1 - \rho_2 &= \rho_1 (\rho_0 - f(-\beta)) + \frac{1}{|\Omega|} \int_{\Omega} [f(-\beta) - f(u^m)] f'(u^m) \\ &\geq \tau_0 (\rho_0 - f(-\beta)) - \frac{1}{\tau_1} \frac{1}{|\Omega|} \int_{\Omega} [f(-\beta) - f(u^m)] = (\rho_0 - f(-\beta)) \left(\tau_0 + \frac{1}{\tau_1} \right). \end{aligned}$$

Based on the above estimates, we deduce that

$$K(\beta) \leq (1 - \delta)\beta + \frac{\tau^2}{2} (f(\beta) - \rho_0) \left(\frac{1}{\tau} + f'(\beta) - \tau_0 - \frac{1}{\tau_1} \right) \leq (1 - \delta)\beta,$$

if τ satisfies $\frac{1}{\tau} + f'(\beta) - \tau_0 - \frac{1}{\tau_1} \geq 0$ or $\tau \leq \left(\frac{1}{\tau_1} + \tau_0 - f'(\beta)\right)^{-1}$. By the same argument, we also have

$$K(-\beta) \geq -(1-\delta)\beta + \frac{\tau^2}{2}(f(-\beta) - \rho_0) \left(\frac{1}{\tau} + f'(-\beta) - \tau_0 - \frac{1}{\tau_1}\right) \geq -(1-\delta)\beta,$$

if $\tau \leq \left(\frac{1}{\tau_1} + \tau_0 - f'(-\beta)\right)^{-1}$. Now we can conclude that $|K(\xi)| \leq (1-\delta)\beta$ for all $\xi \in [-\beta, \beta]$ and therefore,

$$\left\| (1-\delta)u^m + \frac{\tau}{2}\bar{f}(u^m) + \frac{\tau^2}{2}\bar{g}(u^m) \right\| \leq (1-\delta)\beta, \quad (3.11)$$

provided that

$$\tau \leq \min \left\{ \sqrt{\frac{\frac{7}{4} - 2\delta}{\alpha_2(f(-\beta) - f(\beta))}}, \left(\frac{1}{\tau_1} + \tau_0 - f'(\beta)\right)^{-1}, \left(\frac{1}{\tau_1} + \tau_0 - f'(-\beta)\right)^{-1} \right\}.$$

From (3.8), (3.9) and (3.11), we deduce that

$$\|u^{m+1}\| \leq \delta\beta + (1-\delta)\beta = \beta,$$

provided that

$$\tau \leq \min \left\{ 2\delta\tau_1, \sqrt{\frac{\frac{7}{4} - 2\delta}{\alpha_2(f(-\beta) - f(\beta))}}, \left(\frac{1}{\tau_1} + \tau_0 - f'(\pm\beta)\right)^{-1} \right\}.$$

To maximize the upper bound for τ , we pick $\delta = \delta_0$ so that

$$2\delta\tau_1 = \sqrt{\frac{\frac{7}{4} - 2\delta}{\alpha_2(f(-\beta) - f(\beta))}},$$

and the condition on τ becomes $0 < \tau \leq \tau_2$. $\square$

Remark 3.1 The quartic double-well potential function $F(u) = \frac{1}{4}(u^2 - 1)^2$ has the corresponding nonlinear function $f(u) = -F'(u) = u - u^3$. As shown in [32, 60], for any $\beta \in [\frac{2}{\sqrt{3}}, \infty)$, f satisfies Assumptions (A1), (A2) and (A3), hence we choose $\beta = \frac{2}{\sqrt{3}}$. By simple calculations, we obtain

$$\tau_0 = 1, \quad \tau_1 = \frac{1}{3}, \quad \alpha_2 = 4\sqrt{3}, \quad \delta_0 \approx 0.54, \quad \text{and} \quad \tau_2 = \frac{1}{7}.$$

Thus, by Theorems 3.3 and 3.4, the MBP is preserved if $\tau \in (0, \frac{1}{3}]$ for the LRI1 schemes and $\tau \in (0, \frac{1}{7}]$ for the LRI2 scheme. Similarly, the Flory-Huggins potential function

$$F(u) = \frac{\theta}{2} [(1+u) \ln(1+u) + (1-u) \ln(1-u)] - \frac{\theta_c}{2} u^2,$$

corresponds to the nonlinear function $f(u) = -F'(u) = \frac{\theta}{2} \ln \frac{1-u}{1+u} + \theta_c u$, where $0 < \theta < \theta_c$. If $\theta = 0.8$ and $\theta_c = 1.6$, then $\beta \approx 0.987$ (see [32, 60] for a detailed explanation) and so

$$\tau_0 = 0.8, \quad \tau_1 \approx 0.035, \quad \alpha_2 \approx 2290.028, \quad \delta_0 \approx 0.338, \quad \text{and} \quad \tau_2 \approx 0.017.$$

Therefore, the MBP is preserved if $\tau \in (0, 0.035]$ for the LRI1 schemes and $\tau \in (0, 0.017]$ for the LRI2 scheme.

It should be noted that these upper bounds for τ (with either double-well or Flory-Huggins potential function) can be improved in the numerical experiments since the supremum norm of numerical solution is usually less than β . In other words, the above upper bounds for τ are optimal only if the supremum norm is sufficiently close to β .

4 Convergence Analysis

In this section, we assume the nonlinear function f is sufficiently smooth for error analysis purpose; note that the double-well and Flory-Huggins potentials in Remark 3.1 are infinitely differentiable functions in their domains. Throughout the section, Lemma 3.3 is implicitly invoked whenever the exact solution u of (1.1) is present as an argument of the nonlinear functions f , $\bar{f}$ or g , $\bar{g}$. The lemma below shows the Lipschitz continuity of the nonlinear functions $\bar{f}$ and $\bar{g}$.

Lemma 4.1 *If $u, v \in C(\overline{\Omega})$ satisfy $\|u\| \leq \beta$ and $\|v\| \leq \beta$ then*

- (i) $\|\bar{f}(u) - \bar{f}(v)\| \leq 2\alpha_1\|u - v\|$,
- (ii) $\|\bar{g}(u) - \bar{g}(v)\| \leq 4(\alpha_1^2 + \alpha_0\alpha_2)\|u - v\|$.

Proof By the definition of $\bar{f}$, we have

$$\bar{f}(u) - \bar{f}(v) = f(u) - f(v) - \frac{1}{|\Omega|} \int_{\Omega} [f(u) - f(v)].$$

Taking the supremum norm on both sides of the above identity gives us

$$\begin{aligned} \|\bar{f}(u) - \bar{f}(v)\| &\leq \|f(u) - f(v)\| + \frac{1}{|\Omega|} \int_{\Omega} \|f(u) - f(v)\| \\ &= 2\|f(u) - f(v)\| \leq 2\alpha_1\|u - v\|. \end{aligned}$$

On the other hand, we note that $\|\bar{f}(v)\| \leq 2\alpha_0$ and $\|f'(u) - f'(v)\| \leq \alpha_2\|u - v\|$, so

$$\begin{aligned} \|g(u) - g(v)\| &= \|f'(u)\bar{f}(u) - f'(v)\bar{f}(v)\| \\ &\leq \|f'(u)(\bar{f}(u) - \bar{f}(v))\| + \|(f'(u) - f'(v))\bar{f}(v)\| \\ &\leq 2\alpha_1^2\|u - v\| + 2\alpha_0\alpha_2\|u - v\| = 2(\alpha_1^2 + \alpha_0\alpha_2)\|u - v\|. \end{aligned}$$

It follows that

$$\|\bar{g}(u) - \bar{g}(v)\| \leq 2\|g(u) - g(v)\| \leq 4(\alpha_1^2 + \alpha_0\alpha_2)\|u - v\|,$$

which completes the proof. $\square$

From this point forward, we assume that the exact solution u of (1.1) is Lipschitz continuous with respect to the time variable so that $\bar{f}(u)$ and $\bar{g}(u)$ also satisfy the same property. More precisely, there exists a constant $L > 0$ such that

$$\|u(s) - u(t)\| \leq L|s - t|, \quad \forall s, t \in [0, T]. \quad (4.1)$$

By Lemma 4.1, we can assert that

$$\|\bar{f}(u(s)) - \bar{f}(u(t))\| \leq 2\alpha_1 L|s - t|,$$

$$\|\bar{g}(u(s)) - \bar{g}(u(t))\| \leq 4(\alpha_1^2 + \alpha_0\alpha_2)L|s - t|, \quad \forall s, t \in [0, T].$$

The following technical lemma is useful to obtain error estimates for LRI schemes.

Lemma 4.2 *Suppose that $u \in C([0, T], C^2(\bar{\Omega}))$ is the exact solution of (1.1) and satisfies the Lipschitz condition (4.1). For any $s, t \geq 0$ with $s + t \leq T$, it holds*

$$\|e^{s\varepsilon^2\Delta}\bar{f}(u(t)) - \bar{f}(u(t+s))\| \leq Cs,$$

where $C = \varepsilon^2 \max_{t \in [0, T]} \|\Delta f(u(t))\| + 2\alpha_1 L$ is independent of s and t .

Proof By the fundamental theorem of calculus and Lemma 3.2, we deduce that

$$\begin{aligned} \|(e^{s\varepsilon^2\Delta} - 1)\bar{f}(u(t))\| &= \left\| \int_0^s e^{\sigma\varepsilon^2\Delta} \varepsilon^2 \Delta \bar{f}(u(t)) d\sigma \right\| = \left\| \varepsilon^2 \int_0^s e^{\sigma\varepsilon^2\Delta} \Delta f(u(t)) d\sigma \right\| \\ &\leq \varepsilon^2 \int_0^s \|e^{\sigma\varepsilon^2\Delta} \Delta f(u(t))\| d\sigma \leq \varepsilon^2 s \|\Delta f(u(t))\| \leq \varepsilon^2 c_1 s, \end{aligned} \quad (4.2)$$

where $c_1 = \max_{t \in [0, T]} \|\Delta f(u(t))\|$, so c_1 depends on the $C^2([-\beta, \beta])$ -norm of f and the $C([0, T], C^2(\bar{\Omega}))$ -norm of u .

From Lemma 4.1 and the Lipschitz continuity of u with respect to the time variable (4.1), there holds

$$\|\bar{f}(u(t)) - \bar{f}(u(t+s))\| \leq c_2 s, \quad (4.3)$$

where $c_2 = 2\alpha_1 L$. Combining (4.2) and (4.3) gives us

$$\|e^{s\varepsilon^2\Delta}\bar{f}(u(t)) - \bar{f}(u(t+s))\| \leq \|(e^{s\varepsilon^2\Delta} - 1)\bar{f}(u(t))\| + \|\bar{f}(u(t)) - \bar{f}(u(t+s))\| \leq Cs,$$

where $C = \varepsilon^2 c_1 + c_2$. $\square$

Theorem 4.1 *Suppose that the exact solution u to the conservative Allen–Cahn equation (1.1) belongs to $C([0, T], C^2(\bar{\Omega}))$ and satisfies the Lipschitz condition (4.1) with the initial data $u_0 \in C(\bar{\Omega})$ satisfying $\|u_0\| \leq \beta$. Denote by $\{u^m\}_{0 \leq m \leq M}$ the numerical solution generated by the LRI1a scheme (2.3) or the LRI1b scheme (2.4), then there exists a constant $C > 0$ independent of τ such that the following estimate holds for all $\tau \in (0, \tau_1]$:*

$$\|u^m - u(t_m)\| \leq C(e^{2\alpha_1 t_m} - 1)\tau, \quad m = 0, 1, \dots, M.$$

Proof We first consider the LRI1a scheme (2.3). Subtracting (2.1) from (2.3), we obtain

$$\begin{aligned} u^{m+1} - u(t_{m+1}) &= e^{\tau\varepsilon^2\Delta}(u^m - u(t_m)) + \tau e^{\tau\varepsilon^2\Delta}(\bar{f}(u^m) - \bar{f}(u(t_m))) \\ &\quad + \int_0^\tau e^{(\tau-s)\varepsilon^2\Delta} [e^{s\varepsilon^2\Delta}\bar{f}(u(t_m)) - \bar{f}(u(t_m+s))] ds. \end{aligned}$$

Combining the above equation and Lemmas 3.2, 4.1 and 4.2 gives us

$$\|u^{m+1} - u(t_{m+1})\| \leq (1 + 2\alpha_1\tau)\|u^m - u(t_m)\| + \frac{c_3\tau^2}{2},$$

where

$$c_3 = \varepsilon^2 \max_{t \in [0, T]} \|\Delta f(u(t))\| + 2\alpha_1 L, \quad (4.4)$$

is the constant in Lemma 4.2. This implies that

$$\|u^m - u(t_m)\| \leq \frac{c_3\tau}{4\alpha_1} [(1 + 2\alpha_1\tau)^m - 1] \leq \frac{c_3\tau}{4\alpha_1} (e^{2\alpha_1 m\tau} - 1) = C(e^{2\alpha_1 t_m} - 1)\tau,$$

where $C = \frac{c_3}{4\alpha_1}$.

Next, we show that a similar estimate holds for the LRI1b scheme (2.4). Indeed, by subtracting (2.1) from (2.4), we arrive at

$$\begin{aligned} u^{m+1} - u(t_{m+1}) &= e^{\tau\varepsilon^2\Delta}(u^m - u(t_m)) + \tau[\bar{f}(e^{\tau\varepsilon^2\Delta}u^m) - \bar{f}(e^{\tau\varepsilon^2\Delta}u(t_m))] \\ &\quad + \int_0^\tau [\bar{f}(e^{\tau\varepsilon^2\Delta}u(t_m)) - e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(u(t_m+s))]ds. \end{aligned}$$

Based on Lemmas 3.2 and 4.1, we deduce that

$$\begin{aligned} \|u^{m+1} - u(t_{m+1})\| &\leq (1 + 2\alpha_1\tau)\|u^m - u(t_m)\| \\ &\quad + \int_0^\tau \|\bar{f}(e^{\tau\varepsilon^2\Delta}u(t_m)) - e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(u(t_m+s))\|ds. \end{aligned} \quad (4.5)$$

On the other hand, using formula (2.1) and Lemmas 4.1 and 4.2 yields

$$\begin{aligned} &\|\bar{f}(e^{\tau\varepsilon^2\Delta}u(t_m)) - e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(u(t_m+s))\| \\ &\leq \|\bar{f}(e^{\tau\varepsilon^2\Delta}u(t_m)) - \bar{f}(u(t_{m+1}))\| + \|\bar{f}(u(t_{m+1})) - e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(u(t_m+s))\| \\ &\leq 2\alpha_1\|u(t_{m+1}) - e^{\tau\varepsilon^2\Delta}u(t_m)\| + c_3(\tau - s) \\ &= 2\alpha_1\left\|\int_0^\tau e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(u(t_m+s))ds\right\| + c_3(\tau - s) \\ &\leq 2\alpha_1\int_0^\tau \|\bar{f}(u(t_m+s))\|ds + c_3(\tau - s) \leq 4\alpha_0\alpha_1\tau + c_3(\tau - s), \end{aligned} \quad (4.6)$$

where we have used the fact that $\|\bar{f}(u(t_m+s))\| \leq 2\|f(u(t_m+s))\| \leq 2\alpha_0$. From (4.5) and (4.6), we obtain

$$\begin{aligned} \|u^{m+1} - u(t_{m+1})\| &\leq (1 + 2\alpha_1\tau)\|u^m - u(t_m)\| + \int_0^\tau [4\alpha_0\alpha_1\tau + c_3(\tau - s)]ds \\ &= (1 + 2\alpha_1\tau)\|u^m - u(t_m)\| + \left(4\alpha_0\alpha_1 + \frac{c_3}{2}\right)\tau^2. \end{aligned}$$

By the same argument as in the proof for the LRI1a scheme, the desired inequality holds for $C = 2\alpha_0 + \frac{c_3}{4\alpha_1}$. $\square$

Remark 4.1 The constant C in Theorem 4.1 can be explicitly expressed as:

$$C = 2\alpha_0 + \frac{\varepsilon^2}{4\alpha_1} \max_{t \in [0, T]} \|\Delta f(u(t))\| + \frac{1}{2}L,$$

where α_0 and α_1 are defined in (3.5), and L denotes the Lipschitz constant in time of u (cf. (4.1)). Therefore, C does not change significantly with different values of ε . However, both $\max_{t \in [0, T]} \|\Delta f(u(t))\|$ and L , and consequently C , increase when T becomes larger. We shall verify the dependence of C on ε and T in the numerical experiments (cf. Sect. 6).

Theorem 4.2 Suppose that the exact solution u to the conservative Allen–Cahn equation (1.1) belongs to $\mathcal{C}([0, T], \mathcal{C}^4(\overline{\Omega}))$ and satisfies the Lipschitz condition (4.1) with the initial data $u_0 \in \mathcal{C}(\overline{\Omega})$ satisfying $\|u_0\| \leq \beta$. Denote by $\{u^m\}_{0 \leq m \leq M}$ the numerical solution generated

by the LRI2 scheme (2.7), then there exists a constant $C > 0$ independent of τ such that the following estimate holds for all $\tau \in (0, \tau_2]$:

$$\|u^m - u(t_m)\| \leq C(e^{2\tilde{\alpha}_1 t_m} - 1)\tau^2, \quad m = 0, 1, \dots, M,$$

where $\tilde{\alpha}_1 = \alpha_1 + T(\alpha_1^2 + \alpha_0\alpha_2)$.

Proof Subtracting (2.1) from (2.7), we have

$$\begin{aligned} u^{m+1} - u(t_{m+1}) &= e^{\tau\varepsilon^2\Delta}(u^m - u(t_m)) \\ &\quad + \frac{\tau}{2} \left[e^{\tau\varepsilon^2\Delta}(\bar{f}(u^m) - \bar{f}(u(t_m))) + \bar{f}(e^{\tau\varepsilon^2\Delta}u^m) - \bar{f}(e^{\tau\varepsilon^2\Delta}u(t_m)) \right] \\ &\quad + \frac{\tau^2}{2} e^{\tau\varepsilon^2\Delta}(\bar{g}(u^m) - \bar{g}(u(t_m))) + R(t_m), \end{aligned} \quad (4.7)$$

where $R(t_m)$ is defined by

$$\begin{aligned} R(t_m) &= \frac{\tau}{2} [e^{\tau\varepsilon^2\Delta}\bar{f}(u(t_m)) + \bar{f}(e^{\tau\varepsilon^2\Delta}u(t_m))] + \frac{\tau^2}{2} e^{\tau\varepsilon^2\Delta}\bar{g}(u(t_m)) \\ &\quad - \int_0^\tau e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(u(t_m+s)) ds. \end{aligned}$$

Using Lemmas 3.2 and 4.1, we deduce from (4.7) that

$$\|u^{m+1} - u(t_{m+1})\| \leq (1 + 2\alpha_1\tau + 2(\alpha_1^2 + \alpha_0\alpha_2)\tau^2) \|u^m - u(t_m)\| + \|R(t_m)\|. \quad (4.8)$$

Rewrite $R(t_m)$ as $R(t_m) = R_1(t_m) + R_2(t_m)$, where $R_i(t_m)$, $i = 1, 2$, are given by

$$\begin{aligned} R_1(t_m) &= \frac{\tau}{2} [e^{\tau\varepsilon^2\Delta}\bar{f}(u(t_m)) + \bar{f}(e^{\tau\varepsilon^2\Delta}u(t_m))] - \int_0^\tau e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(e^{s\varepsilon^2\Delta}u(t_m)) ds, \\ R_2(t_m) &= \int_0^\tau e^{(\tau-s)\varepsilon^2\Delta}[\bar{f}(e^{s\varepsilon^2\Delta}u(t_m)) - \bar{f}(u(t_m+s)) + s e^{s\varepsilon^2\Delta}\bar{g}(u(t_m))] ds. \end{aligned}$$

Let $\varphi(s) = e^{(\tau-s)\varepsilon^2\Delta}\bar{f}(e^{s\varepsilon^2\Delta}u(t_m))$ with $s \in [0, \tau]$, then $R_1(t_m)$ can be written in terms of φ as follows

$$R_1(t_m) = \frac{\tau}{2} [\varphi(0) + \varphi(s)] - \int_0^\tau \varphi(s) ds = \frac{1}{2} \int_0^\tau s(\tau-s)\varphi''(s) ds.$$

Thus

$$\|R_1(t_m)\| \leq \frac{\tau^3}{12} \sup_{s \in [0, \tau]} \|\varphi''(s)\|. \quad (4.9)$$

By expanding $\varphi''(s)$ using the chain rule, we find that $\sup_{s \in [0, \tau]} \|\varphi''(s)\| \leq 12c_4$, where c_4 depends on $C^4([-\beta, \beta])$ -norm of f and $C([0, T], C^4(\overline{\Omega}))$ -norm of u , but is independent of τ . Therefore, (4.9) gives us

$$\|R_1(t_m)\| \leq c_4\tau^3. \quad (4.10)$$

To estimate $\|R_2(t_m)\|$, we first observe from Lemma 3.2 that

$$\|R_2(t_m)\| \leq \int_0^\tau \|\bar{f}(e^{s\varepsilon^2\Delta}u(t_m)) - \bar{f}(u(t_m+s)) + s e^{s\varepsilon^2\Delta}\bar{g}(u(t_m))\| ds$$

$$\begin{aligned} &\leq \int_0^\tau \|\bar{f}(e^{s\varepsilon^2\Delta}u(t_m)) - \bar{f}(u(t_m+s)) + s\bar{g}(u(t_m))\| ds \\ &\quad + \int_0^\tau s\|(e^{s\varepsilon^2\Delta} - 1)\bar{g}(u(t_m))\| ds. \end{aligned}$$

Similar to (4.2), we have $\|(e^{s\varepsilon^2\Delta} - 1)\bar{g}(u(t_m))\| \leq \varepsilon^2 c_5 s$, where c_5 is given by $c_5 = \max_{t \in [0, T]} \|\Delta g(u(t))\|$ and c_5 depends on $C^3[-\beta, \beta]$ -norm of f and $\mathcal{C}([0, T], \mathcal{C}^2(\bar{\Omega}))$ -norm of u . Consequently,

$$\begin{aligned} \|R_2(t_m)\| &\leq 2 \int_0^\tau \|f(e^{s\varepsilon^2\Delta}u(t_m)) - f(u(t_m+s)) + sg(u(t_m))\| ds + \frac{\varepsilon^2 c_5}{3} \tau^3 \\ &=: 2 \int_0^\tau \|I_1 + I_2 + I_3 + I_4\| ds + \frac{\varepsilon^2 c_5}{3} \tau^3, \end{aligned} \quad (4.11)$$

where I_j with $j = 1, 2, 3, 4$ are given by

$$\begin{aligned} I_1 &= f(u(t_m+s)) - f(e^{s\varepsilon^2\Delta}u(t_m)) - f'(e^{s\varepsilon^2\Delta}u(t_m)) \int_0^s e^{(s-\sigma)\varepsilon^2\Delta} \bar{f}(u(t_m+\sigma)) d\sigma, \\ I_2 &= [f'(e^{s\varepsilon^2\Delta}u(t_m)) - f'(u(t_m))] \int_0^s e^{(s-\sigma)\varepsilon^2\Delta} \bar{f}(u(t_m+\sigma)) d\sigma, \\ I_3 &= f'(u(t_m)) \int_0^s [e^{(s-\sigma)\varepsilon^2\Delta} \bar{f}(u(t_m+\sigma)) - \bar{f}(u(t_m+s))] d\sigma, \\ I_4 &= s[f'(u(t_m))\bar{f}(u(t_m+s)) - g(u(t_m))]. \end{aligned}$$

By Taylor's expansion and formula (2.2), there holds

$$\|I_1\| \leq \frac{\alpha_2}{2} \left\| \int_0^s e^{(s-\sigma)\varepsilon^2\Delta} \bar{f}(u(t_m+\sigma)) d\sigma \right\|^2 \leq \frac{\alpha_2}{2} (2\alpha_0 s)^2 =: c_6 s^2, \quad (4.12)$$

where $c_6 = 2\alpha_0^2\alpha_2$. Based on (4.2), we also have $\|(e^{s\varepsilon^2\Delta} - 1)u(t_m)\| \leq \varepsilon^2 c_7 s$ where $c_7 = \max_{t \in [0, T]} \|\Delta u(t)\|$ depends on $\mathcal{C}([0, T], \mathcal{C}^2(\bar{\Omega}))$ -norm of u . We thus obtain

$$\|I_2\| \leq \alpha_2 \|(e^{s\varepsilon^2\Delta} - 1)u(t_m)\| \int_0^s \|\bar{f}(u(t_m+\sigma))\| d\sigma \leq 2\varepsilon^2 \alpha_0 \alpha_2 c_7 s^2. \quad (4.13)$$

From Lemma 4.2, it holds that

$$\|I_3\| \leq \alpha_1 \int_0^s c_3 (s-\sigma) d\sigma = \frac{\alpha_1 c_3}{2} s^2, \quad (4.14)$$

where c_3 is given by (4.4). Notice that $g = f'\bar{f}$, so we have from Lemma 4.1 that

$$\|I_4\| = s \|f'(u(t_m))[\bar{f}(u(t_m+s)) - \bar{f}(u(t_m))]\| \leq 2\alpha_1^2 L s^2. \quad (4.15)$$

Combining (4.11)–(4.15) and using the triangle inequality lead to

$$\begin{aligned} \|R_2(t_m)\| &\leq 2 \int_0^\tau \left(c_6 s^2 + 2\varepsilon^2 \alpha_0 \alpha_2 c_7 s^2 + \frac{\alpha_1 c_3}{2} s^2 + 2\alpha_1^2 L s^2 \right) ds + \frac{\varepsilon^2 c_5}{3} \tau^3 \\ &= \frac{\varepsilon^2 c_5 + 2c_6 + 4\varepsilon^2 \alpha_0 \alpha_2 c_7 + \alpha_1 c_3 + 4\alpha_1^2 L}{3} \tau^3 =: c_8 \tau^3. \end{aligned} \quad (4.16)$$

From (4.10) and (4.16), we deduce that $\|R(t_m)\| \leq (c_4 + c_8)\tau^3$. Finally, we get from (4.8) that

$$\|u^{m+1} - u(t_{m+1})\| \leq (1 + 2\alpha_1 \tau + 2(\alpha_1^2 + \alpha_0 \alpha_2) \tau^2) \|u^m - u(t_m)\| + (c_4 + c_8) \tau^3$$

$$\leq (1 + 2\tilde{\alpha}_1 \tau) \|u^m - u(t_m)\| + (c_4 + c_8)\tau^3,$$

where $\tilde{\alpha}_1 = \alpha_1 + T(\alpha_1^2 + \alpha_0\alpha_2)$. Using the discrete Gronwall's inequality, we obtain

$$\|u^m - u(t_m)\| \leq \frac{c_4 + c_8}{2\tilde{\alpha}_1} (e^{2\tilde{\alpha}_1 t_m} - 1)\tau^2.$$

The proof is completed by setting $C = \frac{c_4 + c_8}{2\tilde{\alpha}_1}$. $\square$

Remark 4.2 (i) The constant C in Theorem 4.2 can be written as

$$C = \frac{1}{2\tilde{\alpha}_1} c_4 + \frac{1}{6\tilde{\alpha}_1} \left(\varepsilon^2 \max_{t \in [0, T]} \|\Delta g(u(t))\| + 4\alpha_0^2 \alpha_2 + 4\varepsilon^2 \alpha_0 \alpha_2 \max_{t \in [0, T]} \|\Delta u(t)\| \right. \\ \left. + \varepsilon^2 \alpha_1 \max_{t \in [0, T]} \|\Delta f(u(t))\| + 6\alpha_1^2 L \right).$$

Thus, as for the LRI1 schemes (cf. Remark 4.1), the error constant C for the LRI2 scheme does not vary considerably with respect to ε ; however, it increases when the final time T becomes larger (cf. Sect. 6).

(ii) The temporal error analysis of the LRI1 and LRI2 schemes relies on the Duhamel's formula (2.1) and the Lipschitz condition (4.1), so that the exact solution u is only assumed to be Lipschitz continuous in time for both first- and second-order schemes (cf. Theorems 4.1 and 4.2). Therefore, the proposed schemes are called low regularity integrators [11, 39], in contrast to classical methods such as ETD [32] and isIFRK [60] which require higher regularity in time on the exact solution for the convergence analysis (i.e., u needs to be in C^p in time for the p -order scheme to converge).

5 Energy Stability

Throughout this section, we assume that the spatial domain Ω is either a finite interval ($d = 1$), a rectangle ($d = 2$), or a rectangular parallelepiped ($d = 3$). We first state two preliminary lemmas.

Lemma 5.1 *Suppose that Δ is equipped with the periodic boundary conditions on Ω and $v \in C^1(\overline{\Omega})$ satisfies the same boundary conditions, then $\partial_{x_i}(e^{t\Delta}v) = e^{t\Delta}v_{x_i}$ for all $i = 1, 2, \dots, d$ and $t > 0$, where $v_{x_i} := \partial_{x_i}v$.*

Proof For simplicity, we only consider the one dimensional case, the corresponding proof in higher dimensions can be done in the same way. Without loss of generality, we may assume that $\Omega = (-P, P)$ for some $P > 0$. Let $W(\cdot, t) = e^{t\Delta}v$ and $w(\cdot, t) = e^{t\Delta}v'$, then W and w satisfy the following heat equations:

$$\begin{cases} W_t = W_{xx}, & x \in (-P, P), t > 0, \\ W(x, 0) = v(x), & x \in (-P, P), \end{cases}$$

and

$$\begin{cases} w_t = w_{xx}, & x \in (-P, P), t > 0, \\ w(x, 0) = v'(x), & x \in (-P, P), \end{cases}$$

respectively, subject to the periodic boundary conditions. The general solutions to the two problems above are:

$$W(x, t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{P}\right) + b_n \sin\left(\frac{n\pi x}{P}\right) \right) e^{-(\frac{n\pi}{P})^2 t},$$

$$w(x, t) = \frac{\bar{a}_0}{2} + \sum_{n=1}^{\infty} \left(\bar{a}_n \cos\left(\frac{n\pi x}{P}\right) + \bar{b}_n \sin\left(\frac{n\pi x}{P}\right) \right) e^{-(\frac{n\pi}{P})^2 t},$$

where $a_n, \bar{a}_n, n \geq 0$ and $b_n, \bar{b}_n, n \geq 1$ are given by

$$a_n = \frac{1}{P} \int_{-P}^P v(x) \cos\left(\frac{n\pi x}{P}\right) dx, \quad b_n = \frac{1}{P} \int_{-P}^P v(x) \sin\left(\frac{n\pi x}{P}\right) dx,$$

$$\bar{a}_n = \frac{1}{P} \int_{-P}^P v'(x) \cos\left(\frac{n\pi x}{P}\right) dx, \quad \bar{b}_n = \frac{1}{P} \int_{-P}^P v'(x) \sin\left(\frac{n\pi x}{P}\right) dx.$$

We observe that

$$\partial_x W(x, t) = \sum_{n=1}^{\infty} \frac{n\pi}{P} \left(-a_n \sin\left(\frac{n\pi x}{P}\right) + b_n \cos\left(\frac{n\pi x}{P}\right) \right) e^{-(\frac{n\pi}{P})^2 t} = w(x, t),$$

since $\bar{a}_0 = \frac{1}{P}[v(P) - v(-P)] = 0$ and $\bar{a}_n = \frac{n\pi}{P} b_n, \bar{b}_n = -\frac{n\pi}{P} a_n$ for all $n \geq 1$. The identity in Lemma 5.1 can thus be obtained by taking into account $W(\cdot, t) = e^{t\Delta} v$ and $w(\cdot, t) = e^{t\Delta} v'$. $\square$

Remark 5.1 The two operators ∂_{x_i} and $e^{t\Delta}$ need not be commutative when Δ is equipped with other types of boundary conditions on general domains. Our proof for the energy stability relies on Lemma 5.1, which requires the periodic boundary conditions; it remains an open question to show the same result, i.e., Theorem 5.1, for the homogeneous Neumann boundary conditions.

Lemma 5.2 Let $\{u^m\}_{0 \leq m \leq M}$ be the numerical solution sequence generated by the LRI1a or LRI1b scheme (resp. the LRI2 scheme) and suppose that the exact solution u to (1.1) satisfies the conditions stated in Theorem 4.1 (resp. Theorem 4.2). If $u_0 \in C^1(\bar{\Omega})$ with $\|u_0\| \leq \beta$, then there exist constants c_9 and c_{10} independent of τ such that

- (i) $\|u_{x_i}^m\| \leq c_9$ for all $i = 1, 2, \dots, d$ and $m = 0, 1, \dots, M$.
- (ii) $\|u_{x_i}^{m+1} - u_{x_i}^m\| \leq c_{10}\tau$ for all $i = 1, 2, \dots, d$ and $m = 0, 1, \dots, M-1$.

Proof (i) We first consider the LRI1a scheme (2.3). Taking the derivative on both sides of (2.3) with respect to x_i yields

$$u_{x_i}^{m+1} = e^{\tau \varepsilon^2 \Delta} (u_{x_i}^m + \tau f'(u^m) u_{x_i}^m), \quad (5.1)$$

where we have used Lemma 5.1 and the fact that $\partial_{x_i} \bar{f}(u^m) = \partial_{x_i} f(u^m)$. From Lemma 3.2 and the boundedness of u^m implied by Theorem 3.3, we obtain

$$\|u_{x_i}^{m+1}\| \leq \|u_{x_i}^m + \tau f'(u^m) u_{x_i}^m\| \leq (1 + \alpha_1 \tau) \|u_{x_i}^m\|.$$

Similarly, we observe from the LRI1b scheme (2.4) that

$$\|u_{x_i}^{m+1}\| = \|e^{\tau \varepsilon^2 \Delta} u_{x_i}^m + \tau f'(e^{\tau \varepsilon^2 \Delta} u^m) e^{\tau \varepsilon^2 \Delta} u_{x_i}^m\| \leq (1 + \alpha_1 \tau) \|u_{x_i}^m\|.$$

This gives us $\|u_{x_i}^m\| \leq (1 + \alpha_1 \tau)^m \|u_{x_i}^0\| \leq e^{\alpha_1 t_m} \|u_{x_i}^0\| \leq e^{\alpha_1 T} \|\partial_{x_i} u_0\|$ for all $i = 1, 2, \dots, d$. The result follows by choosing $c_9 = e^{\alpha_1 T} \max_{1 \leq i \leq d} \|\partial_{x_i} u_0\|$.

For the LRI2 scheme (2.7), based on the above estimates we have that

$$\|u_{x_i}^{m+1}\| \leq (1 + \alpha_1 \tau) \|u_{x_i}^m\| + \left\| \partial_{x_i} \left(\frac{\tau^2}{2} e^{\tau \varepsilon^2 \Delta} \bar{g}(u^m) \right) \right\|. \quad (5.2)$$

Using Lemmas 3.2 and 5.1, we notice that

$$\begin{aligned} \left\| \partial_{x_i} \left(\frac{\tau^2}{2} e^{\tau \varepsilon^2 \Delta} \bar{g}(u^m) \right) \right\| &= \left\| \frac{\tau^2}{2} e^{\tau \varepsilon^2 \Delta} (\partial_{x_i} g(u^m)) \right\| \leq \frac{\tau^2}{2} \|\partial_{x_i} (f'(u^m) \bar{f}(u^m))\| \\ &= \frac{\tau^2}{2} \|[f'(u^m))^2 + \bar{f}(u^m) f''(u^m)] u_{x_i}^m\| \\ &\leq \frac{\tau^2}{2} (\alpha_1^2 + 2\alpha_0 \alpha_2) \|u_{x_i}^m\|. \end{aligned} \quad (5.3)$$

From (5.2) and (5.3), we deduce that

$$\|u_{x_i}^{m+1}\| \leq \left[1 + \alpha_1 \tau + \left(\frac{\alpha_1^2}{2} + \alpha_0 \alpha_2 \right) \tau^2 \right] \|u_{x_i}^m\|.$$

The assertion for the LRI2 scheme follows by the same argument as above.

(ii) It should be noted that for any $u_1, u_2, v_1, v_2 \in \mathcal{C}(\bar{\Omega})$ with $\|u_i\|, \|v_i\| \leq \beta$ for $i = 1, 2$, we have

$$\begin{aligned} \|f'(u_1)v_1 - f'(u_2)v_2\| &\leq \|f'(u_1)(v_1 - v_2)\| + \|(f'(u_1) - f'(u_2))v_2\| \\ &\leq \alpha_1 \|v_1 - v_2\| + \alpha_2 \|v_2\| \|u_1 - u_2\|. \end{aligned} \quad (5.4)$$

Theorems 4.1 and 4.2 allow us to have the following estimate

$$\begin{aligned} \|u^{m+1} - u^m\| &\leq \|u^{m+1} - u(t_{m+1})\| + \|u(t_{m+1}) - u(t_m)\| + \|u(t_m) - u^m\| \\ &= C\tau + L\tau + C\tau = (2C + L)\tau. \end{aligned} \quad (5.5)$$

For the LRI1a scheme (2.3), applying (5.1) with m and $m - 1$ and taking the difference between the resulting equations, we get

$$u_{x_i}^{m+1} - u_{x_i}^m = e^{\tau \varepsilon^2 \Delta} [u_{x_i}^m - u_{x_i}^{m-1} + \tau (f'(u^m) u_{x_i}^m - f'(u^{m-1}) u_{x_i}^{m-1})].$$

The estimates (5.4) and (5.5), together with the result in part (i), lead to

$$\begin{aligned} \|u_{x_i}^{m+1} - u_{x_i}^m\| &\leq \|u_{x_i}^m - u_{x_i}^{m-1}\| + \tau \|f'(u^m) u_{x_i}^m - f'(u^{m-1}) u_{x_i}^{m-1}\| \\ &\leq (1 + \alpha_1 \tau) \|u_{x_i}^m - u_{x_i}^{m-1}\| + \alpha_2 c_9 (2C + L) \tau^2, \end{aligned}$$

which claims the assertion by the discrete Gronwall's inequality.

Similarly, we also have from the LRI1b scheme (2.4) that

$$\begin{aligned} \|u_{x_i}^{m+1} - u_{x_i}^m\| &= \left\| e^{\tau \varepsilon^2 \Delta} (u_{x_i}^m - u_{x_i}^{m-1}) + \tau [f'(e^{\tau \varepsilon^2 \Delta} u^m) e^{\tau \varepsilon^2 \Delta} u_{x_i}^m - f'(e^{\tau \varepsilon^2 \Delta} u^{m-1}) e^{\tau \varepsilon^2 \Delta} u_{x_i}^{m-1}] \right\| \\ &\leq \|u_{x_i}^m - u_{x_i}^{m-1}\| + \tau \|f'(e^{\tau \varepsilon^2 \Delta} u^m) e^{\tau \varepsilon^2 \Delta} u_{x_i}^m - f'(e^{\tau \varepsilon^2 \Delta} u^{m-1}) e^{\tau \varepsilon^2 \Delta} u_{x_i}^{m-1}\| \\ &\leq (1 + \alpha_1 \tau) \|u_{x_i}^m - u_{x_i}^{m-1}\| + \alpha_2 c_9 (2C + L) \tau^2, \end{aligned}$$

here we have used (5.4) and (5.5) in the last estimate.

Using above estimates and (5.3) for the LRI2 scheme (2.7), we have

$$\begin{aligned} \|u_{x_i}^{m+1} - u_{x_i}^m\| &\leq (1 + \alpha_1 \tau) \|u_{x_i}^m - u_{x_i}^{m-1}\| + \alpha_2 c_9 (2C + L) \tau^2 \\ &\quad + \left\| \partial_{x_i} \left(\frac{\tau^2}{2} e^{\tau \varepsilon^2 \Delta} \bar{g}(u^m) \right) - \partial_{x_i} \left(\frac{\tau^2}{2} e^{\tau \varepsilon^2 \Delta} \bar{g}(u^{m-1}) \right) \right\| \\ &\leq (1 + \alpha_1 \tau) \|u_{x_i}^m - u_{x_i}^{m-1}\| + [\alpha_2 c_9 (2C + L) + c_9 (\alpha_1^2 + 2\alpha_0 \alpha_2)] \tau^2, \end{aligned}$$

which finishes the proof. $\square$

Theorem 5.1 Assume that the initial data $u_0 \in C^1(\overline{\Omega})$ with $\|u_0\| \leq \beta$ and the exact solution u of (1.1) satisfies the conditions stated in Theorem 4.1 (resp. Theorem 4.2), then the numerical solution $\{u^m\}_{0 \leq m \leq M}$ generated by the LRI1a or LRI1b scheme (resp. the LRI2 scheme) satisfies

$$E(u^m) \leq E(u_0) + C,$$

where $C > 0$ is independent of τ .

Proof By the definition of E , it follows immediately that

$$E(u^{m+1}) - E(u^m) = \int_{\Omega} \left[\frac{\varepsilon^2}{2} (|\nabla u^{m+1}|^2 - |\nabla u^m|^2) + F(u^{m+1}) - F(u^m) \right] dx. \quad (5.6)$$

Note that $F' = -f$ and $\|u^{m+1} - u^m\| \leq (2C + L)\tau$ by (5.5), hence

$$\int_{\Omega} [F(u^{m+1}) - F(u^m)] dx \leq \alpha_0 |\Omega| \|u^{m+1} - u^m\| \leq \alpha_0 |\Omega| (2C + L) \tau. \quad (5.7)$$

On the other hand, it is implied from Lemma 5.2 that

$$\begin{aligned} \int_{\Omega} \frac{\varepsilon^2}{2} (|\nabla u^{m+1}|^2 - |\nabla u^m|^2) dx &= \frac{\varepsilon^2}{2} \sum_{i=1}^d \int_{\Omega} (u_{x_i}^{m+1} - u_{x_i}^m)(u_{x_i}^{m+1} + u_{x_i}^m) dx \\ &\leq \frac{\varepsilon^2}{2} \sum_{i=1}^d \int_{\Omega} \|u_{x_i}^{m+1} - u_{x_i}^m\| (\|u_{x_i}^{m+1}\| + \|u_{x_i}^m\|) dx \\ &\leq \varepsilon^2 d c_9 c_{10} |\Omega| \tau. \end{aligned} \quad (5.8)$$

Combining (5.6), (5.7) and (5.8) leads to

$$E(u^{m+1}) - E(u^m) \leq [\alpha_0 (2C + L) + \varepsilon^2 d c_9 c_{10}] |\Omega| \tau,$$

which is the desired estimate. $\square$

6 Numerical Experiments

We carry out several numerical examples to investigate the convergence of the proposed LRI schemes and three important properties of the discrete solution, namely mass conservation, the MBP and energy stability. While there are various types of spatial discretization that can be coupled with LRIs, we use the central finite difference method to take advantage of FFT-based fast implementation [13, 26, 32, 61]. In all test cases below, the periodic boundary conditions are considered over the spatial domain $\Omega = (-0.5, 0.5)^2$ in two dimensions (2D) and $\Omega = (-0.5, 0.5)^3$ in three dimensions (3D); similar results can be obtained with the

Table 1 L^2 errors and convergence rates by the first-order ETD and LRI schemes with the double-well potential and $(\varepsilon, T) \in \{(0.01, 1), (0.005, 2), (0.0025, 4)\}$

ε	τ	ETD1		LRI1a		LRI1b	
		L^2 error	Rate	L^2 error	Rate	L^2 error	Rate
0.01	$T/16$	4.86e−03		4.98e−03		4.85e−03	
	$T/32$	2.46e−03	0.99	2.52e−03	0.98	2.45e−03	0.99
	$T/64$	1.23e−03	0.99	1.27e−03	0.99	1.23e−03	0.99
	$T/128$	6.19e−04	1.00	6.35e−04	1.00	6.17e−04	1.00
	$T/256$	3.10e−04	1.00	3.18e−04	1.00	3.09e−04	1.00
0.005	$T/16$	1.79e−02		1.81e−02		1.78e−02	
	$T/32$	9.08e−03	0.98	9.21e−03	0.98	9.03e−03	0.98
	$T/64$	4.58e−03	0.99	4.64e−03	0.99	4.55e−03	0.99
	$T/128$	2.30e−03	0.99	2.33e−03	0.99	2.28e−03	0.99
	$T/256$	1.15e−03	1.00	1.17e−03	1.00	1.14e−03	1.00
0.0025	$T/16$	3.83e−02		3.93e−02		3.77e−02	
	$T/32$	1.93e−02	0.99	1.99e−02	0.98	1.88e−02	1.00
	$T/64$	9.64e−03	1.00	1.00e−02	0.99	9.40e−03	1.00
	$T/128$	4.82e−03	1.00	5.01e−03	1.00	4.69e−03	1.00
	$T/256$	2.41e−03	1.00	2.51e−03	1.00	2.34e−03	1.00

homogeneous Neumann boundary conditions. The nonlinear function is either one of the following two types:

$$f_1(u) = u - u^3, \quad \text{and} \quad f_2(u) = \frac{\theta}{2} \ln \frac{1-u}{1+u} + \theta_c u,$$

corresponding to the double-well and Flory-Huggins potential functions, respectively. For the latter, we choose $\theta = 0.8$ and $\theta_c = 1.6$.

6.1 Convergence Tests

We consider the 2D conservative Allen–Cahn equation (1.1) with the initial data given by

$$u_0(x, y) = \frac{1}{2} \cos(2\pi x) \cos(2\pi y), \quad (x, y) \in \overline{\Omega},$$

so that $\|u_0\| \leq \beta$, where $\beta \in \{2/\sqrt{3}, 0.987\}$ is a uniform bound mentioned in Remark 3.1. The spatial mesh size $h = h_x = h_y = 1/2048$ is fixed, where h_x and h_y denote the mesh sizes in x and y directions, respectively.

For the polynomial potential function $f = f_1$, the interfacial parameter ε is chosen to be 0.01, 0.005 or 0.0025 and the final time T is given by 1, 2 or 4, respectively. We vary the time step size $\tau \in \{T/16, T/32, \dots, T/256\}$ and compute both L^2 and L^∞ errors of the numerical solutions produced by first and second-order LRI schemes at the final time step. The reference solution is obtained using the LRI2 scheme with a refined time step size $\tau = T/1024$. To investigate the performance of the LRI schemes, especially LRI2, we compare them with the ETD methods of the same order. The L^2 and L^∞ errors and convergence rates of the first-order ETD and LRI schemes are presented in Tables 1 and 2, respectively. We clearly observe

Table 2 L^∞ errors and convergence rates by the first-order ETD and LRI schemes with the double-well potential and $(\varepsilon, T) \in \{(0.01, 1), (0.005, 2), (0.0025, 4)\}$

ε	τ	ETD1		LRI1a		LRI1b	
		L^∞ error	Rate	L^∞ error	Rate	L^∞ error	Rate
0.01	$T/16$	7.77e-03		7.92e-03		7.79e-03	
	$T/32$	3.93e-03	0.98	4.01e-03	0.98	3.94e-03	0.98
	$T/64$	1.98e-03	0.99	2.02e-03	0.99	1.98e-03	0.99
	$T/128$	9.93e-04	1.00	1.01e-03	1.00	9.94e-04	1.00
	$T/256$	4.97e-04	1.00	5.07e-04	1.00	4.98e-04	1.00
0.005	$T/16$	3.50e-02		3.53e-02		3.50e-02	
	$T/32$	1.80e-02	0.96	1.81e-02	0.96	1.79e-02	0.96
	$T/64$	9.10e-03	0.98	9.17e-03	0.98	9.09e-03	0.98
	$T/128$	4.58e-03	0.99	4.61e-03	0.99	4.57e-03	0.99
	$T/256$	2.30e-03	1.00	2.31e-03	1.00	2.29e-03	1.00
0.0025	$T/16$	1.47e-01		1.47e-01		1.47e-01	
	$T/32$	7.75e-02	0.92	7.80e-02	0.92	7.74e-02	0.92
	$T/64$	3.98e-02	0.96	4.01e-02	0.96	3.97e-02	0.96
	$T/128$	2.02e-02	0.98	2.03e-02	0.98	2.01e-02	0.98
	$T/256$	1.01e-02	0.99	1.02e-02	0.99	1.01e-02	0.99

the first-order convergence of the ETD1, LRI1a and LRI1b schemes, and the errors increase as ε gets closer to zero and T becomes larger simultaneously. Both tables show that the errors produced by the ETD1 scheme are slightly smaller than those generated by the LRI1a scheme, but slightly larger than those produced by the LRI1b scheme. This observation leads to the numerical conclusion that the LRI1b scheme is the best among the three first-order schemes. For the second-order case, the L^2 and L^∞ errors and the corresponding convergence rates by ETD1 and LRI2 are reported in Table 3, where the difference between these two schemes becomes clearer. As ε goes to zero, L^2 errors by the LRI2 scheme are nearly half of those by the ETD1 scheme, while L^∞ errors by the LRI2 scheme are roughly three quarters of those by its counterpart. Thus, LRI2 performs much better than ETD1, especially when ε approaches zero; besides, the second-order accuracy are well observed for both schemes.

For the Flory-Huggins potential function $f = f_2$, we choose $\varepsilon = 0.0025$ or 0.001 and fix $T = 4$ for simplicity. The L^2 and L^∞ errors produced by the first-order schemes, i.e., ETD1, LRI1a and LRI1b, are presented in Tables 4 and 5, respectively. As expected, we observe the first-order convergence of all three schemes with a quite similar performance in terms of error magnitude. Table 6 shows the second-order convergence of both ETD1 and LRI2 schemes. However, L^2 errors by LRI2 are approximately half of those by ETD1, while L^∞ errors by LRI2 are about two thirds of those by its counterpart. Since T is fixed in this case, the errors produced by our first- and second-order LRI schemes do not change significantly with different values of ε (cf. Remarks 4.1 and 4.2(i)). We refer to [11] for the running time comparison of the LRI and ETD methods, where LRIs require less computational time compared to the ETD schemes of the same order.

Table 3 Errors and convergence rates by the second-order ETD and LRI schemes with the double-well potential and $(\varepsilon, T) \in \{(0.01, 1), (0.005, 2), (0.0025, 4)\}$

ε	τ	ETDRK2			LRI2			
		L^2 error	Rate	L^∞ error	Rate	L^2 error	Rate	L^∞ error
0.01	$T/16$	1.41e-04		1.72e-04		1.34e-04		2.33e-04
	$T/32$	3.56e-05	1.98	4.39e-05	1.97	3.38e-05	1.99	5.85e-05
	$T/64$	8.93e-06	2.00	1.11e-05	1.99	8.46e-06	2.00	1.46e-05
	$T/128$	2.21e-06	2.01	2.75e-06	2.01	2.09e-06	2.01	3.61e-06
	$T/256$	5.28e-07	2.07	6.56e-07	2.07	4.99e-07	2.07	8.60e-07
0.005	$T/16$	1.07e-03		1.49e-03		5.55e-04		9.85e-04
	$T/32$	2.69e-04	1.98	3.90e-04	1.94	1.41e-04	1.98	2.55e-04
	$T/64$	6.76e-05	1.99	9.94e-05	1.97	3.53e-05	1.99	6.47e-05
	$T/128$	1.68e-05	2.01	2.48e-05	2.00	8.77e-06	2.01	1.61e-05
	$T/256$	4.00e-06	2.07	5.95e-06	2.06	2.09e-06	2.07	3.86e-06
0.0025	$T/16$	3.97e-03		1.19e-02		1.97e-03		9.17e-03
	$T/32$	1.02e-03	1.96	3.25e-03	1.87	5.28e-04	1.90	2.46e-03
	$T/64$	2.59e-04	1.97	8.46e-04	1.94	1.37e-04	1.95	6.36e-04
	$T/128$	6.48e-05	2.00	2.14e-04	1.98	3.45e-05	1.99	1.60e-04
	$T/256$	1.55e-05	2.06	5.15e-05	2.05	8.29e-06	2.06	3.85e-05

Table 4 L^2 errors and convergence rates by the first-order ETD and LRI schemes with the Flory-Huggins potential and $T = 4$

ε	τ	ETD1		LRI1a		LRI1b	
		L^2 error	Rate	L^2 error	Rate	L^2 error	Rate
0.0025	$T/16$	4.45e-02		4.47e-02		4.42e-02	
	$T/32$	2.24e-02	0.99	2.27e-02	0.98	2.23e-02	0.99
	$T/64$	1.13e-02	0.99	1.15e-02	0.99	1.12e-02	0.99
	$T/128$	5.67e-03	0.99	5.75e-03	0.99	5.62e-03	1.00
	$T/256$	2.84e-03	1.00	2.88e-03	1.00	2.81e-03	1.00
0.001	$T/16$	4.56e-02		4.56e-02		4.56e-02	
	$T/32$	2.30e-02	0.99	2.31e-02	0.98	2.30e-02	0.99
	$T/64$	1.16e-02	0.98	1.17e-02	0.98	1.16e-02	0.99
	$T/128$	5.85e-03	0.99	5.86e-03	0.99	5.84e-03	0.99
	$T/256$	2.93e-03	1.00	2.94e-03	1.00	2.93e-03	1.00

6.2 Mass Conservation, MBP and Energy Stability Tests

We now consider both 2D and 3D conservative Allen–Cahn equation (1.1) with a random initial data ranging from -0.9 and 0.9 . In the 2D problem with the double-well potential, we test all LRI schemes in terms of mass conservation, MBP and energy stability. For other test cases, the LRI2 scheme is used due to its high accuracy.

Table 5 L^∞ errors and convergence rates by the first-order ETD and LRI schemes with the Flory-Huggins potential and $T = 4$

ε	τ	ETD1		LRI1a		LRI1b	
		L^∞ error	Rate	L^∞ error	Rate	L^∞ error	Rate
0.0025	$T/16$	1.37e-01		1.37e-01		1.37e-01	
	$T/32$	7.19e-02	0.93	7.21e-02	0.93	7.19e-02	0.93
	$T/64$	3.68e-02	0.97	3.69e-02	0.97	3.67e-02	0.97
	$T/128$	1.86e-02	0.98	1.87e-02	0.98	1.86e-02	0.98
	$T/256$	9.34e-03	0.99	9.38e-03	0.99	9.33e-03	0.99
0.001	$T/16$	1.38e-01		1.38e-01		1.38e-01	
	$T/32$	7.25e-02	0.93	7.26e-02	0.93	7.25e-02	0.93
	$T/64$	3.71e-02	0.97	3.71e-02	0.97	3.71e-02	0.97
	$T/128$	1.88e-02	0.98	1.88e-02	0.98	1.88e-02	0.98
	$T/256$	9.43e-03	0.99	9.43e-03	0.99	9.43e-03	0.99

Table 6 Errors and convergence rates by the second-order ETD and LRI schemes with the Flory-Huggins potential and $T = 4$

ε	τ	ETDRK2				LRI2			
		L^2 error	Rate	L^∞ error	Rate	L^2 error	Rate	L^∞ error	Rate
0.0025	$T/16$	1.50e-02		2.49e-02		3.01e-03		6.97e-03	
	$T/32$	1.21e-03	3.63	3.55e-03	2.81	5.57e-04	2.43	1.84e-03	1.92
	$T/64$	2.77e-04	2.13	7.61e-04	2.22	1.39e-04	2.00	4.72e-04	1.96
	$T/128$	6.69e-05	2.05	1.79e-04	2.09	3.45e-05	2.01	1.18e-04	2.00
	$T/256$	1.58e-05	2.08	4.19e-05	2.10	8.26e-06	2.06	2.84e-05	2.06
0.001	$T/16$	1.53e-02		2.55e-02		3.07e-03		6.99e-03	
	$T/32$	1.25e-03	3.61	3.64e-03	2.81	5.61e-04	2.45	1.84e-03	1.92
	$T/64$	2.84e-04	2.14	7.78e-04	2.23	1.39e-04	2.01	4.72e-04	1.96
	$T/128$	6.84e-05	2.05	1.82e-04	2.09	3.45e-05	2.01	1.18e-04	2.00
	$T/256$	1.61e-05	2.08	4.26e-05	2.10	8.24e-06	2.07	2.84e-05	2.06

6.2.1 Two-Dimensional Coarsening

We first simulate long time phase separation processes of the model equation in 2D with $\varepsilon = 0.01$ and the double-well potential function. Snapshots of the numerical solution at different times $t = 1, 10, 20, 40, 100$ and 300 are depicted in Fig. 1; the results are obtained by the LRI2 scheme with the time step size $\tau = 0.1$ and the spatial mesh size $h = h_x = h_y = 1/1024$.

The evolution of the absolute mass error (the difference between the mass along the time and the initial mass), the supremum norm and the discrete energy of the first- and second-order LRI schemes with $\tau \in \{0.1, 0.2, 0.5\}$ and $T = 100$ is described in Figs. 2, 3 and 4. We see that LRIs preserve mass and MBP, and decrease the energy functional over time. Since the supremum norm curves eventually reach values around 1, which is strictly less than $\beta = 2/\sqrt{3}$, we are allowed to take $\tau = 0.5$ in this case.

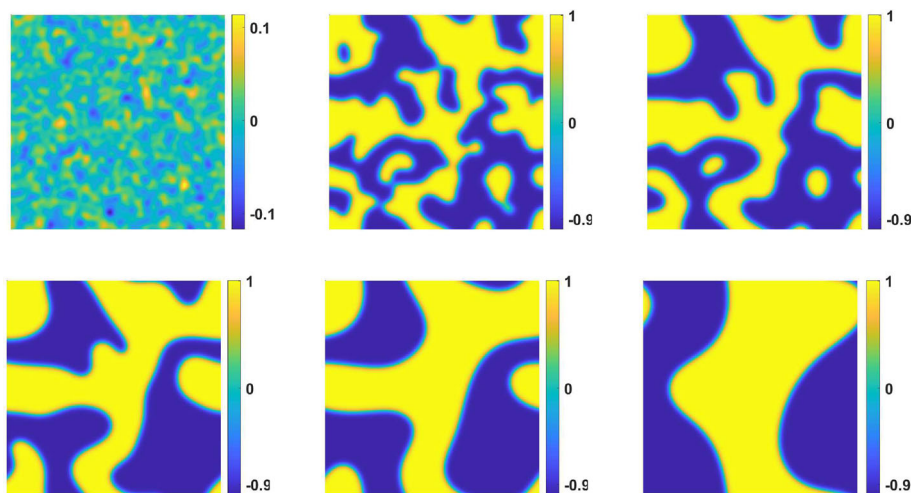


Fig. 1 Snapshots of the numerical solution at $t = 1, 10, 20, 40, 100$ and 300 (from left to right and top to bottom) produced by the LRI2 scheme with $\tau = 0.1$ and the double-well potential

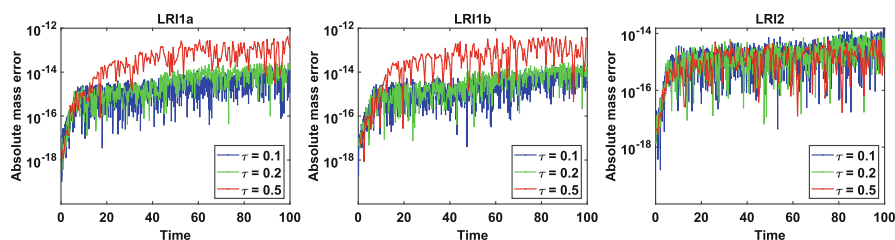


Fig. 2 Evolution of the absolute mass error of the numerical solutions by the LRI schemes with $\tau = 0.1, 0.2, 0.5$ and the double-well potential

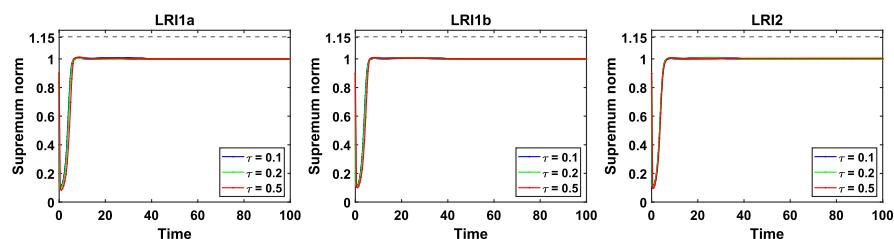


Fig. 3 Evolution of the supremum norm of the numerical solutions by the LRI schemes with $\tau = 0.1, 0.2, 0.5$ and the double-well potential

We now perform the simulations with the Flory-Huggins potential. Figure 5 shows snapshots of the numerical solution by LRI2 with $\tau = 0.1$ at time $t = 1, 10, 20, 40, 100$ and 300 . The corresponding absolute mass error, supremum norm and energy of the numerical solutions with $\tau \in \{0.1, 0.2\}$ are illustrated in Fig. 6. All expected properties are again satisfied. It should be noted that $\tau = 0.5$ does not work in this case because the condition on τ is more restrictive than the case of double-well potential (cf. Remark 3.1).

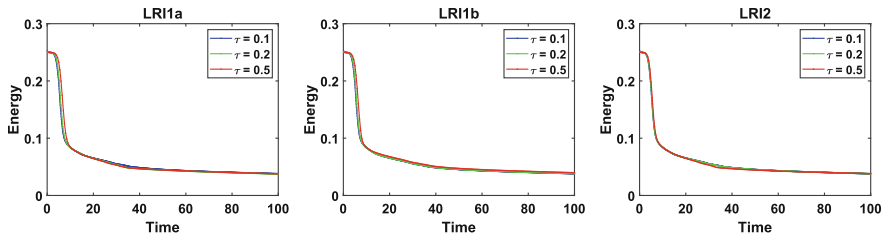


Fig. 4 Evolution of the energy of the numerical solutions by the LRI schemes with $\tau = 0.1, 0.2, 0.5$ and the double-well potential

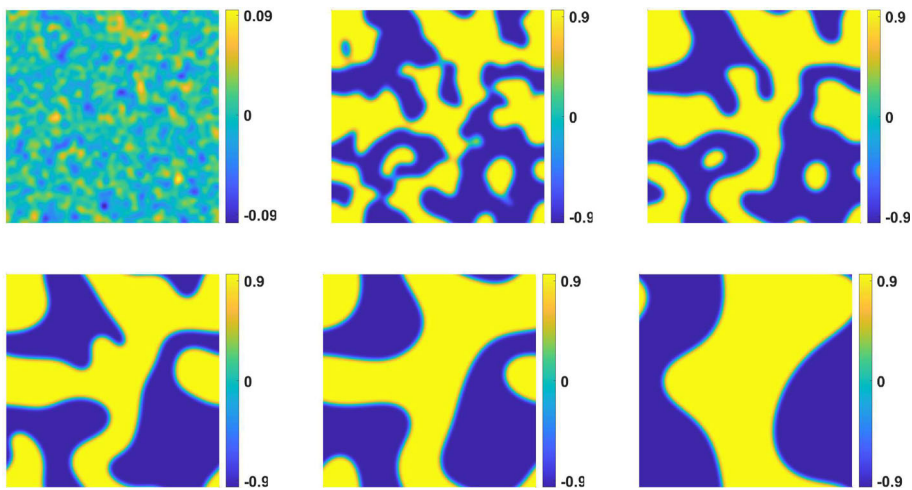


Fig. 5 Snapshots of the numerical solution at $t = 1, 10, 20, 40, 100$ and 300 (from left to right and top to bottom) produced by the LRI2 scheme with $\tau = 0.1$ and the Flory-Huggins potential

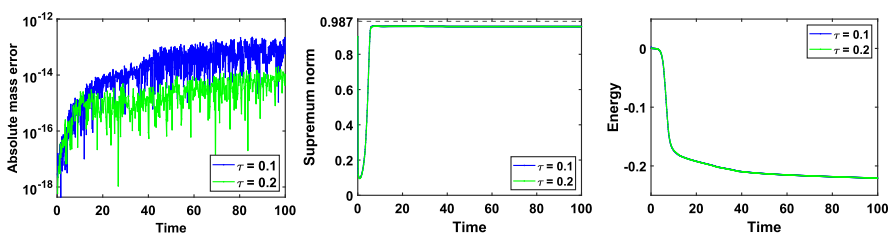


Fig. 6 Evolution of the absolute mass error (left), the supremum norm (middle) and the energy (right) of the numerical solutions by the LRI2 scheme with $\tau = 0.1, 0.2$ and the Flory-Huggins potential

6.2.2 Three-Dimensional Coarsening

We perform 3D simulations of the conservative Allen–Cahn equation with fixed $\varepsilon = 0.01$ and spatial mesh size $h = h_x = h_y = h_z = 1/128$. For the double-well potential, the evolution of 3D isosurfaces computed by the LRI2 scheme with $\tau = 0.1$ at $t = 1, 40, 100$ and 300 is presented in Fig. 7. Figure 8 demonstrates mass conservation, the MBP property and the energy decay of the LRI2 scheme with different time step sizes $\tau \in \{0.1, 0.2, 0.5\}$.

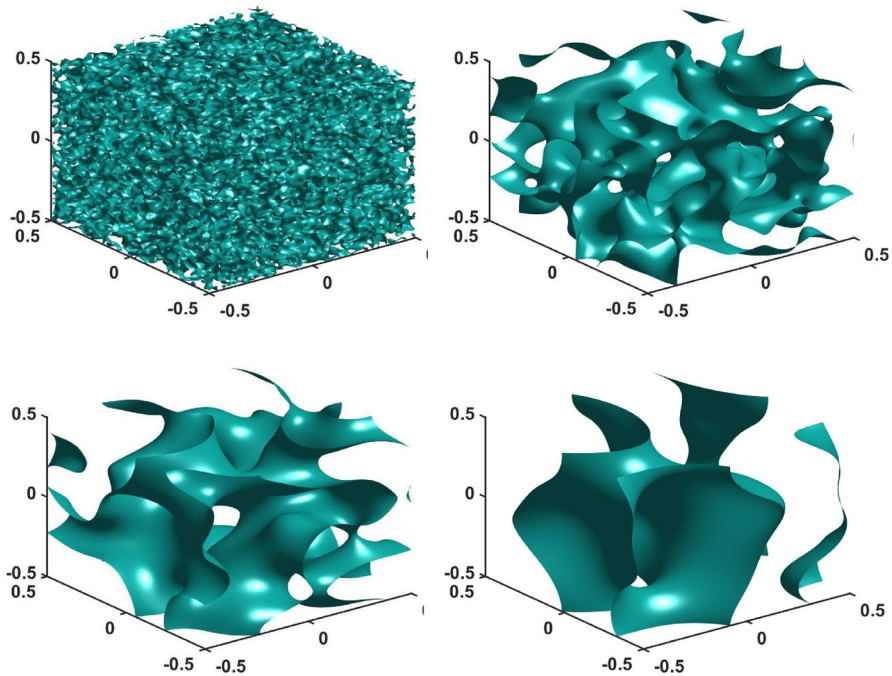


Fig. 7 Isosurface snapshots of the numerical solution at $t = 1, 40, 100$ and 300 (from left to right and top to bottom) produced by the LRI2 scheme with $\tau = 0.1$ and the double-well potential

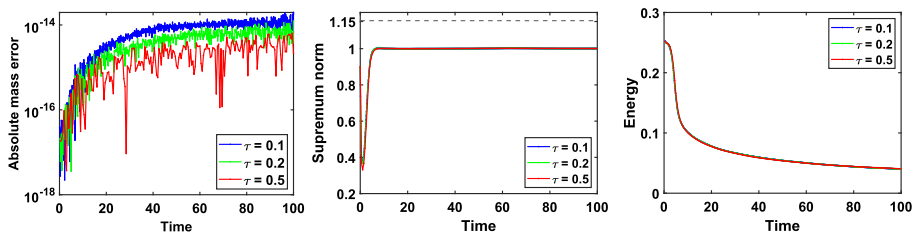


Fig. 8 Evolution of the absolute mass error (left), the supremum norm (middle) and the energy (right) of the numerical solutions by the LRI2 scheme with $\tau = 0.1, 0.2, 0.5$ and the double-well potential in 3D

For the Flory-Huggins potential, isosurface snapshots by the LRI2 scheme with $\tau = 0.1$ at $t = 1, 40, 100$ and 300 are shown in Fig. 9. All three properties are verified in Fig. 10 with $\tau = 0.1$ and $\tau = 0.2$.

7 Conclusions

In this work, we designed and analyzed the first- and second-order LRI schemes for the conservative Allen–Cahn equation with a nonlocal, space-independent Lagrange multiplier. The proposed schemes were shown to unconditionally preserve the total mass over time. Different from the classical Allen–Cahn equation, the presence of the nonlocal Lagrange multiplier

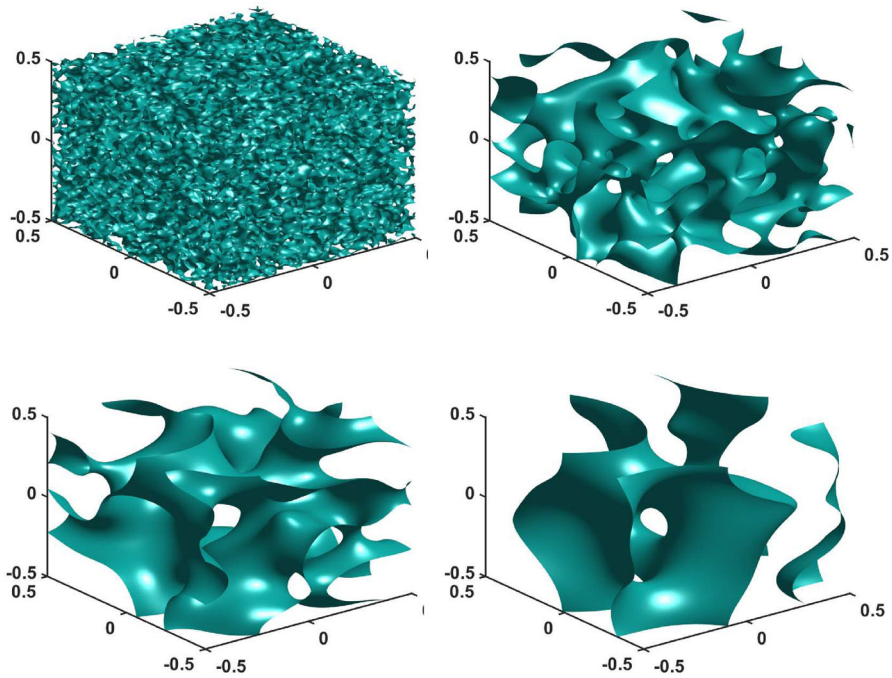


Fig. 9 Isosurface snapshots of the numerical solution at $t = 1, 40, 100$ and 300 (from left to right and top to bottom) produced by the LRI2 scheme with $\tau = 0.1$ and the Flory-Huggins potential

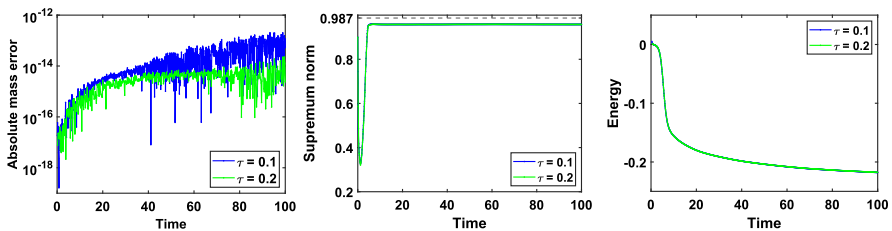


Fig. 10 Evolution of the absolute mass error (left), the supremum norm (middle) and the energy (right) of the numerical solutions by the LRI2 scheme with $\tau = 0.1, 0.2$ and the Flory-Huggins potential in 3D

imposed more restrictive constraints on the time step size for LRIs to theoretically satisfy the MBP. Convergence analysis was derived based on the MBP results at both continuous and discrete levels under the assumption that the exact solution of the model equation is Lipschitz continuous in time. By combining the MBP property and error estimates, we established energy stability for the LRI schemes on rectangular domains with the periodic boundary conditions. To confirm the theoretical findings, we carried out numerical experiments with different test cases and showed the outperformance of LRIs compared to the ETD schemes, as well as their preservation of mass, MBP and energy stability. The results also suggested that the upper bounds for the time step size can be improved, as the numerical solutions never reach the constant β in Assumption (A1). Future research includes further investigations on energy stability of LRIs for the conservative Allen–Cahn equation on general domains with the homogeneous Neumann boundary conditions, the application of LRIs to other types of

mass-conserving Allen–Cahn equations [3, 25] as well as the development of high-order LRIs for the Navier–Stokes equations [33] and the coupled Navier–Stokes/Allen–Cahn system [24].

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Declarations

Conflict of interest The authors have not disclosed any Conflict of interest.

References

1. Aihara, S., Takaki, T., Takada, N.: Multi-phase-field modeling using a conservative Allen–Cahn equation for multiphase flow. *Comput. Fluids* **178**, 141–151 (2019)
2. Allen, S.M., Cahn, J.W.: A microscopic theory for antiphase boundary motion and its application to antiphase domain coarsening. *Acta Metall.* **27**(6), 1085–1095 (1979)
3. Brassel, M., Bretin, E.: A modified phase field approximation for mean curvature flow with conservation of the volume. *Math. Meth. Appl. Sci.* **34**(10), 1157–1180 (2011)
4. Calvo, M.C., Schratz, K.: Uniformly accurate low regularity integrators for the Klein–Gordon equation from the classical to nonrelativistic limit regime. *SIAM J. Numer. Anal.* **60**(2), 888–912 (2022)
5. Chen, X., Hilhorst, D., Logak, E.: Mass conserving Allen–Cahn equation and volume preserving mean curvature flow. *Interf. Free Bound.* **12**(4), 527–549 (2011)
6. Cheng, Q., Wang, C.: Error estimate of a second order accurate scalar auxiliary variable (SAV) numerical method for the epitaxial thin film equation. *Adv. Appl. Math. Mech.* **13**, 1318–1354 (2021)
7. Choi, Y., Kim, J.: Maximum principle preserving and unconditionally stable scheme for a conservative Allen–Cahn equation. *Eng. Anal. Bound. Elem.* **150**, 111–119 (2023)
8. Choi, J.W., Lee, H.G., Jeong, D., Kim, J.: An unconditionally gradient stable numerical method for solving the Allen–Cahn equation. *Phys. A* **388**(9), 1791–1803 (2009)
9. Copetti, M.I.M., Elliott, C.M.: Numerical analysis of the Cahn–Hilliard equation with a logarithmic free energy. *Numer. Math.* **63**(1), 39–65 (1992)
10. Debuissche, A., Dettori, L.: On the Cahn–Hilliard equation with a logarithmic free energy. *Nonlinear Anal. Theory Methods Appl.* **24**(10), 1491–1514 (1995)
11. Doan, C.-K., Hoang, T.-T.-P., Ju, L., Schratz, K.: Low regularity integrators for semilinear parabolic equations with maximum bound principles. *BIT Numer. Math.* **63**(1), 2 (2023)
12. Doan, C.-K., Hoang, T.-T.-P., Ju, L.: Fully discrete error analysis of first-order low regularity integrators for the Allen–Cahn equation. *Numer. Methods Partial Differ. Equ.* **39**(5), 3594–3608 (2023)
13. Du, Q., Ju, L., Li, X., Qiao, Z.: Maximum principle preserving exponential time differencing schemes for the nonlocal Allen–Cahn equation. *SIAM J. Numer. Anal.* **57**(2), 875–898 (2019)
14. Du, Q., Ju, L., Li, X., Qiao, Z.: Maximum bound principles for a class of semilinear parabolic equations and exponential time-differencing schemes. *SIAM Rev.* **63**(2), 317–359 (2021)
15. Feng, X., Prohl, A.: Numerical analysis of the Allen–Cahn equation and approximation for mean curvature flows. *Numer. Math.* **94**(1), 33–65 (2003)
16. Feng, J., Zhou, Y., Hou, T.: A maximum-principle preserving and unconditionally energy-stable linear second-order finite difference scheme for Allen–Cahn equations. *Appl. Math. Lett.* **118**, 107179 (2021)
17. Geng, S., Li, T., Ye, Q., Yang, X.: A new conservative Allen–Cahn type Ohta–Kawasaki phase-field model for diblock copolymers and its numerical approximations. *Adv. Appl. Math. Mech.* **14**(1), 101–124 (2022)
18. He, D., Pan, K., Hu, H.: A spatial fourth-order maximum principle preserving operator splitting scheme for the multi-dimensional fractional Allen–Cahn equation. *Appl. Numer. Math.* **151**, 44–63 (2020)
19. Hou, T., Leng, H.: Numerical analysis of a stabilized Crank–Nicolson/Adams–Bashforth finite difference scheme for Allen–Cahn equations. *Appl. Math. Lett.* **102**, 106150 (2020)
20. Hou, T., Tang, T., Yang, J.: Numerical analysis of fully discretized Crank–Nicolson scheme for fractional-in-space Allen–Cahn equations. *J. Sci. Comput.* **72**(3), 1214–1231 (2017)
21. Hou, T., Xiu, D., Jiang, W.: A new second-order maximum-principle preserving finite difference scheme for Allen–Cahn equations with periodic boundary conditions. *Appl. Math. Lett.* **104**, 106265 (2020)

22. Huang, Z., Lin, G., Ardekani, A.M.: Consistent and conservative scheme for incompressible two-phase flows using the conservative Allen-Cahn model. *J. Comput. Phys.* **420**, 109718 (2020)
23. Huang, Z., Lin, G., Ardekani, A.M.: A consistent and conservative volume distribution algorithm and its applications to multiphase flows using phase-field models. *Int. J. Multiph. Flow* **142**, 103727 (2021)
24. Jeong, D., Kim, J.: Conservative Allen-Cahn-Navier-Stokes system for incompressible two-phase fluid flows. *Comput. Fluids* **156**, 239–246 (2017)
25. Jiang, K., Ju, L., Li, J., Li, X.: Unconditionally stable exponential time differencing schemes for the mass-conserving Allen-Cahn equation with nonlocal and local effects. *Numer. Methods Partial Differ. Equ.* **38**(6), 1636–1657 (2022)
26. Ju, L., Zhang, J., Zhu, L., Du, Q.: Fast explicit integration factor methods for semilinear parabolic equations. *J. Sci. Comput.* **62**(2), 431–455 (2015)
27. Kim, J., Jeong, D., Yang, S.D., Choi, Y.: A finite difference method for a conservative Allen-Cahn equation on non-flat surfaces. *J. Comput. Phys.* **334**, 170–181 (2017)
28. Kim, J., Lee, S., Choi, Y.: A conservative Allen-Cahn equation with a space-time dependent Lagrange multiplier. *Int. J. Eng. Sci.* **84**, 11–17 (2014)
29. Lee, H.G.: High-order and mass conservative methods for the conservative Allen-Cahn equation. *Comput. Math. Appl.* **72**(3), 620–631 (2016)
30. Lee, D., Kim, J.: Comparison study of the conservative Allen-Cahn and the Cahn-Hilliard equations. *Math. Comput. Simul.* **119**, 35–56 (2016)
31. Lee, H.G., Shin, J., Lee, J.Y.: A high-order and unconditionally energy stable scheme for the conservative Allen-Cahn equation with a nonlocal Lagrange multiplier. *J. Sci. Comput.* **90**, 1–12 (2022)
32. Li, J., Ju, L., Cai, Y., Feng, X.: Unconditionally maximum bound principle preserving linear schemes for the conservative Allen-Cahn equation with nonlocal constraint. *J. Sci. Comput.* **87**(3), 1–32 (2021)
33. Li, B., Ma, S., Schratz, K.: A semi-implicit exponential low-regularity integrator for the Navier-Stokes equations. *SIAM J. Numer. Anal.* **60**(4), 2273–2292 (2022)
34. Liao, H.L., Tang, T., Zhou, T.: A second-order and nonuniform time-stepping maximum-principle preserving scheme for time-fractional Allen-Cahn equations. *J. Comput. Phys.* **414**, 109473 (2020)
35. Ostermann, A., Rousset, F., Schratz, K.: Error estimates of a Fourier integrator for the cubic Schrödinger equation at low regularity. *Found. Comput. Math.* **21**(3), 725–765 (2021)
36. Ostermann, A., Schratz, K.: Low regularity exponential-type integrators for semilinear Schrödinger equations. *Found. Comput. Math.* **18**, 731–755 (2018)
37. Protter, M.H., Weinberger, H.F.: *Maximum Principles in Differential Equations*. Springer, New York (1984)
38. Ren, F., Song, B., Sukop, M.C., Hu, H.: Improved lattice Boltzmann modeling of binary flow based on the conservative Allen-Cahn equation. *Phys. Rev. E* **94**(2), 023311 (2016)
39. Rousset, F., Schratz, K.: A general framework of low regularity integrators. *SIAM J. Numer. Anal.* **59**(3), 1735–1768 (2021)
40. Rubinstein, J., Sternberg, P.: Nonlocal reaction-diffusion equations and nucleation. *IMA J. Appl. Math.* **48**(3), 249–264 (1992)
41. Schratz, K., Wang, Y., Zhao, X.: Low-regularity integrators for nonlinear Dirac equations. *Math. Comput.* **90**(327), 189–214 (2021)
42. Shen, J., Xu, J., Yang, J.: A new class of efficient and robust energy stable schemes for gradient flows. *SIAM Rev.* **61**, 474–506 (2019)
43. Shen, J., Tang, T., Yang, J.: On the maximum principle preserving schemes for the generalized Allen-Cahn equation. *Commun. Math. Sci.* **14**(6), 1517–1534 (2016)
44. Shen, J., Yang, X.: Numerical approximations of Allen-Cahn and Cahn-Hilliard equations. *Discrete Contin. Dyn. Syst* **28**(4), 1669–1691 (2010)
45. Shin, J., Lee, H.G., Lee, J.Y.: Unconditionally stable methods for gradient flow using convex splitting Runge-Kutta scheme. *J. Comput. Phys.* **347**, 367–381 (2017)
46. Tan, Z., Zhang, C.: The discrete maximum principle and energy stability of a new second-order difference scheme for Allen-Cahn equations. *Appl. Numer. Math.* **166**, 227–237 (2021)
47. Tang, T., Yang, J.: Implicit-explicit scheme for the Allen-Cahn equation preserves the maximum principle. *J. Comput. Math.* **34**(5), 471–481 (2016)
48. Wang, C., Wise, S.M.: An energy stable and convergent finite-difference scheme for the modified phase field crystal equation. *SIAM J. Numer. Anal.* **49**(3), 945–969 (2011)
49. Wu, J., Yang, J., Tan, Z.: Unconditionally energy-stable time-marching methods for the multi-phase conservative Allen-Cahn fluid models based on a modified SAV approach. *Comput. Meth. Appl. Mech. Eng.* **398**, 115291 (2022)
50. Wu, Y., Zhao, X.: Optimal convergence of a second-order low-regularity integrator for the KdV equation. *IMA J. Numer. Anal.* **42**(4), 3499–3528 (2022)

51. Wu, Y., Zhao, X.: Embedded exponential-type low-regularity integrators for KdV equation under rough data. *BIT Numer. Math.* **62**(3), 1049–1090 (2022)
52. Xiao, X., He, R., Feng, X.: Unconditionally maximum principle preserving finite element schemes for the surface Allen-Cahn type equations. *Numer. Methods Partial Differ. Equ.* **36**(2), 418–438 (2020)
53. Xu, J., Li, Y., Wu, S., Bousquet, A.: On the stability and accuracy of partially and fully implicit schemes for phase field modeling. *Comput. Methods Appl. Mech. Eng.* **345**, 826–853 (2019)
54. Xu, Z., Yang, X., Zhang, H., Xie, Z.: Efficient and linear schemes for anisotropic Cahn-Hilliard model using the stabilized-invariant energy quadratization (S-IEQ) approach. *Comput. Phys. Commun.* **238**, 36–49 (2019)
55. Yang, J., Jeong, D., Kim, J.: A fast and practical adaptive finite difference method for the conservative Allen-Cahn model in two-phase flow system. *Int. J. Multiph. Flow* **137**, 103561 (2021)
56. Yang, J., Kim, J.: Numerical study of incompressible binary fluids on 3D curved surfaces based on the conservative Allen-Cahn-Navier-Stokes model. *Comput. Fluids* **228**, 105094 (2021)
57. Yang, X., Zhang, G.: Convergence analysis for the invariant energy quadratization (IEQ) schemes for solving the Cahn-Hilliard and Allen-Cahn equations with general nonlinear potential. *J. Sci. Comput.* **82**(3), 55 (2020)
58. Zhai, S., Weng, Z., Feng, X.: Investigations on several numerical methods for the non-local Allen-Cahn equation. *Int. J. Heat Mass Transfer* **87**, 11118 (2015)
59. Zhai, S., Weng, Z., Feng, X.: Fast explicit operator splitting method and time-step adaptivity for fractional non-local Allen-Cahn model. *Appl. Math. Model.* **40**(2), 1315–1324 (2016)
60. Zhang, H., Yan, J., Qian, X., Chen, X., Song, S.: Explicit third-order unconditionally structure-preserving schemes for conservative Allen-Cahn equations. *J. Sci. Comput.* **90**, 1–29 (2022)
61. Zhang, H., Yan, J., Qian, X., Song, S.: Numerical analysis and applications of explicit high order maximum principle preserving integrating factor Runge-Kutta schemes for Allen-Cahn equation. *Appl. Numer. Math.* **161**, 372–390 (2021)

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